Online Appendix for "Timing Search"[1]

Contents

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A Theory and Microfoundations

A.1 One-Sided Timing Search

The one-sided timing search problem considers an agent who observes market conditions at Poisson rate λ and chooses when to act. With i.i.d. payoffs, the reservation value satisfies the standard McCall equation

$$rv^* = -c + \lambda \int_{v^*}^{\bar{v}} (v - v^*)\, dF(v),$$

and timing search is isomorphic to cross-sectional search.

Under Ornstein–Uhlenbeck payoffs ($dv = \theta(\bar{v} - v)dt + \sigma\, dW$), two results emerge with no cross-sectional analog. First, the reservation value depends on mean-reversion speed θ: $\partial v^*/\partial\theta < 0$ when $v^* > \bar{v}$ and > 0 when $v^* < \bar{v}$. Second, the option value satisfies $\frac{\partial}{\partial v}[W(v) - v] < -1$ for $v \gg \bar{v}$ (agents rush to act at high payoffs), versus exactly -1 under i.i.d. search.

A.2 Matching Microfoundation for Gaussian Profits

This appendix provides a matching-theoretic microfoundation for the Gaussian matching efficiency function $m(n) = \exp(-\frac{\varphi}{2}(n - n_p)^2)$ stated in Assumption 2 of the main text (Gaussian scale efficiency).

Consider a market with n active firms. Each potential match between a firm and a trading partner has a quality θ drawn from a normal distribution $\theta \sim N(n_p, 1/\varphi)$. A firm operating at market thickness n captures matches with quality near n: specifically, matches within distance ε of n on the quality dimension. The density of good matches available to the firm is proportional to the normal density evaluated at n:

$$m(n) \propto \frac{1}{\sqrt{2\pi/\varphi}} \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big)\,.$$

This adapts the spatial-competition logic of Hotelling (1929) and Salop (1979) to market thickness: firms compete for matches along a continuum (the quality-density dimension) rather than along a geographic line or circle, and the normal distribution generates the Gaussian kernel. The connection to the equal-value structure of timing search runs through Burdett and Mortensen (1998), who carried the spatial-equilibrium logic into wage posting; the present paper carries the same logic into the time dimension via the ironing condition. Absorbing the normalizing constant into π_0 yields $\pi(n) = \pi_0 \exp(-\frac{\varphi}{2}(n - n_p)^2)$.

The parameter φ inherits its interpretation from the variance of match qualities: $\varphi = 1/\mathrm{Var}(\theta)$. Industries with homogeneous match qualities (low variance, high φ) have a narrow peak and are sensitive to scale deviations. Industries with heterogeneous qualities (high variance, low φ) have a broad peak and are robust to deviations.

A.3 Derivation of the Timing Friction

This appendix derives the exponential-quadratic timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ as the continuous-time limit of proportional adjustment costs.

Discrete-time formulation Consider a discrete-time economy with period length $\Delta > 0$. In each period, effective output is reduced by a fraction proportional to the square of the adjustment speed:

$$y_t = \pi(n_t) \cdot \left[1 - \tfrac{\mu}{2}\left(\frac{n_{t+\Delta} - n_t}{\Delta}\right)^2 \Delta\right].$$

The adjustment cost is proportional to the square of the rate of change $(\Delta n/\Delta)^2$, scaled by the period length Δ. The parameter $\mu > 0$ governs the severity of the friction.

Continuous-time limit As $\Delta \to 0$, write $\dot{n} = \lim_{\Delta \to 0}(n_{t+\Delta} - n_t)/\Delta$ and note that $1 - x \approx e^{-x}$ for small x. The effective output becomes:

$$y(t) = \pi(n(t)) \cdot e^{-\frac{\mu}{2}\dot{n}(t)^2} = \pi(n(t)) \cdot \xi(\dot{n}(t)).$$

Robustness of the quadratic form Any symmetric friction $\xi(\dot{n})$ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$, where $\mu = -\xi''(0)$. The leading-order dynamics—in particular $\omega = \sqrt{\varphi/\mu}$—depend only on μ, not on higher-order terms. The exponential-quadratic form is therefore canonical.

A.4 Detailed Derivation of Entry-Matching Micro-Foundation

This appendix provides the detailed derivation of Assumption 3 in the main text (micro-foundation of the timing friction from entry-exit matching congestion).

Setup The matching function from the main text (§2.3) is $m(S, R) = S^{\alpha} R^{1-\alpha}$, where $S = M - n$ is the searcher pool and R is the stock of entry resources. The matching probability per searcher is $p(\theta_e) = \theta_e^{-(1-\alpha)}$ with $\theta_e = S/R$. The matching infrastructure R is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$.

Steady state At $\dot{n} = 0$, the realized flow equals $e_{ss} = \delta n_p$, the infrastructure operates at its tuned capacity, and the matching probability equals $p_{ss} = p(\theta_e^{ss})$.

Bottleneck strain The gross entry flow is $e(t) = \dot{n}(t) + \delta n(t)$, with steady-state $e_{ss} = \delta n_p$, so the deviation of gross entry from the steady-state level is $e - e_{ss} = \dot{n} + \delta(n - n_p)$. Following the modeling convention adopted in the main text (Section 2.4), we attribute the slow-moving replacement component $\delta n(t)$ to a baseline capacity that tracks the prevailing population $n(t)$ and adjusts smoothly without bottleneck strain. The bottleneck-strain pipeline services only the net adjustment pressure $\dot{n}(t)$, which is the part of gross entry that exceeds the prevailing replacement rate. When $\dot{n} \neq 0$, this adjustment pressure deviates from zero in either direction. Rapid entry ($\dot{n} > 0$) strains the entry pipeline (permits, construction, specialist labor). Rapid net exit ($\dot{n} < 0$) strains the reallocation pipeline (resale markets, write-down procedures, workforce reassignment). Both reduce the matching probability per searcher below p_{ss}. We do not adopt a single-bottleneck specification in which one pipeline would service the full $|e - e_{ss}| = |\dot{n} + \delta(n - n_p)|$; the maintained model defines the bottleneck margin as net adjustment pressure $\dot{n}$ alone, in keeping with the modeling convention of Section 2.4 of the main text.

We represent this strain by a CES-like penalty that enters multiplicatively into the matching probability:

$$p(\theta_e; \dot{n}) = \theta_e^{-(1-\alpha)} \cdot \left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right]^{-(1-\alpha)/\alpha}. \tag{1}$$

The strain term depends on $\dot{n}^2$ rather than on signed $\dot{n}$, encoding the symmetry between entry and exit congestion. The $(\delta n_p)^2$ scale in the denominator reflects that infrastructure capacity is calibrated to steady-state throughput: deviations small relative to δn_p are handled easily, while deviations approaching δn_p in magnitude strain the system to saturation.

What is assumed versus what is derived The specific CES functional form in (1) is a modeling choice: any smooth, symmetric strain penalty $g\big(\dot{n}^2/(\delta n_p)^2\big)$ with $g(0) = 0$ and $g'(0) = 1/2$ yields the same second-order behavior. The universality result in Online Appendix C shows that *any* smooth symmetric friction ξ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$ with $\mu = -\xi''(0)$; the second-order coefficient is therefore invariant to the particular smoothing chosen. What the specific form in (1) delivers is not the quadratic *shape* of the friction—that shape is universal—but the specific *scaling formula*

$$\mu = \frac{1-\alpha}{\alpha(\delta n_p)^2},$$

which pins the constant in terms of the matching elasticity α and the steady-state throughput δn_p. This scaling follows from two observations that hold for any reasonable smoothing: (i) the only natural scale for $\dot{n}$ is δn_p (steady-state throughput), so μ must be inversely proportional to $(\delta n_p)^2$ by dimensional reasoning; (ii) the matching elasticity α enters through the Cobb-Douglas term $\theta_e^{-(1-\alpha)}$, producing the $(1-\alpha)/\alpha$ factor regardless of which specific strain function is chosen. The CES form is a convenient representative of this equivalence class; alternatives (e.g., Gaussian strain $\exp[-\dot{n}^2/(2(\delta n_p)^2)]$, or quadratic strain $[1 - \dot{n}^2/(2(\delta n_p)^2)]$) yield the identical μ at second order.

Effective friction The effective output friction from the entry-replacement bottleneck is:

$$\ln \xi(\dot{n}) \equiv \ln p(\theta_e;\, \dot{n}) - \ln p(\theta_e^{ss}) = -\frac{1-\alpha}{\alpha} \ln\left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right].$$

Taylor expansion For small $|\dot{n}|/(\delta n_p)$, $\ln(1 + x) = x - x^2/2 + \cdots$, and only the leading term contributes at second order in $\dot{n}$:

$$\ln \xi(\dot{n}) \approx -\frac{1-\alpha}{\alpha} \cdot \frac{\dot{n}^2}{2(\delta n_p)^2} + O(\dot{n}^4).$$

This is $-\frac{\mu}{2}\dot{n}^2$ with

$$\mu = \frac{1-\alpha}{\alpha(\delta n_p)^2} > 0.$$

Verification The friction ξ is symmetric ($\xi(\dot{n}) = \xi(-\dot{n})$), correctly signed ($\xi < 1$ for $\dot{n} \neq 0$), smooth (all derivatives exist at $\dot{n} = 0$), and matches the canonical quadratic form of Online Appendix C to leading order. For $\alpha = 0.5$ (symmetric matching), $\mu = 1/(\delta n_p)^2$. For $\alpha = 0.7$ (searcher-weighted), $\mu = 3/(7(\delta n_p)^2) \approx 0.43/(\delta n_p)^2$, approximately 57% smaller—implying faster cycles in industries with higher searcher matching power.

B Proofs of Main-Text Results

B.1 Proof of Proposition 2 (Stochastic Limit Cycle)

This appendix provides the proof of Proposition 2 in the main text, which states that the spectral density of $n(t)$ under small stochastic forcing peaks at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ in the undamped small-shock limit; with bottleneck damping $\eta = \chi/\mu > 0$, the observed PSD peak shifts to $\nu_{\text{peak}} = \sqrt{\varphi/\mu - \eta^2/2}$, equal to ω_0 up to an $O(\eta^2)$ correction.

Economic interpretation of the ironing condition The ironing-condition energy identity $\frac{\mu}{2}\dot{n}^2 + \frac{\varphi}{2}(n - n_p)^2 = E$ of the main text decomposes into two economically interpretable pieces. The term $\frac{\mu}{2}\dot{n}^2$ is the per-firm flow cost of operating the matching kernel $\xi(\dot{n})$ at rate $\dot{n}$—the cost of keeping the entry rate at its current pace. The term $\frac{\varphi}{2}(n - n_p)^2$ is the per-firm profit shortfall from operating at scale n rather than at the optimum n_p. The ironing condition pins the sum of these two quantities at a constant level $E > 0$ along the equilibrium orbit: this is the equal-value level required by indifference across all entry dates. The four-phase cycle of Figure C.4 traces the periodic exchange between the two pieces: at $n = n_p$ the entire E sits in the entry-rate flow cost ($\dot{n}$ at maximum); at the turning points $n = n_p \pm A$ the entire E sits in the profit shortfall ($\dot{n} = 0$); in between, the two trade against each other while their sum remains pinned. The damped equilibrium adds the bottleneck dissipation channel $\chi\dot{n}^2$, which slowly erodes the equal-value level along the orbit and is balanced in the stationary equilibrium by injection from productivity shocks $\sigma_\varepsilon\varepsilon_t$.

Action–angle coordinates Write the deterministic solution as $n(t) = n_p + A\cos\phi(t)$ and $\dot{n}(t) = -A\omega\sin\phi(t)$, where $\phi(t) = \omega t + \phi_0$ and $\omega = \sqrt{\varphi/\mu}$. Transform the stochastic system to action–angle coordinates (A, ϕ).

Under the perturbation $\tilde{\pi} = \pi + \sigma_\varepsilon \varepsilon_t$, the equilibrium value level $E = \frac{\varphi}{2}A^2$ evolves as:

$$\mathrm{d}E = \frac{\sigma_\varepsilon \varepsilon_t}{\pi(n)} \dot{n}\,\mathrm{d}t.$$

Since $\dot{n} = -A\omega\sin\phi$ and $\pi(n) \approx \pi_0 e^E$ near the equilibrium orbit:

$$\mathrm{d}A \approx -\frac{\sigma_\varepsilon}{\varphi A \pi_0 e^E}\, A\omega \sin\phi\, \varepsilon_t\, \mathrm{d}t.$$

Averaging over the fast phase ϕ, the amplitude A evolves as a diffusion with drift coefficient determined by the noise coupling.

Spectral density The process $n(t)$ is a stochastically forced damped harmonic oscillator:

$$\ddot{n} + \eta\,\dot{n} + \omega^2(n - n_p) = \text{noise},$$

where the noise term inherits the i.i.d. structure of ε_t. The transfer function of this damped oscillator is:

$$H(\nu) = \frac{1}{\omega^2 - \nu^2 + i\eta\nu},$$

where $\eta = \chi/\mu$ is the bottleneck damping rate (with χ the bottleneck friction coefficient and μ the matching friction). The power spectral density is $S(\nu) = |H(\nu)|^2 S_\varepsilon(\nu)$. Maximizing $|H(\nu)|^2 = 1/[(\omega^2 - \nu^2)^2 + \eta^2\nu^2]$ over ν yields the driven-system peak at $\nu^2_{\text{peak}} = \omega^2 - \eta^2/2$, i.e. slightly below the natural frequency $\omega = \sqrt{\varphi/\mu}$. Since S_ε is flat (white noise), the peak location is determined entirely by (ω, η), not by σ_ε.

Properties

(i) For $\eta \ll \omega$ (under-damped regime), the spectral peak is approximately at $\omega = \sqrt{\varphi/\mu}$ with a correction of order η^2/ω that is negligible at empirically relevant damping levels.

(ii) The peak location depends on (φ, μ, η) only; the shock variance σ^2_ε scales the peak *height* without shifting its location or its normalized width (which is governed by η).

(iii) The persistence of the cycle is endogenous: even i.i.d. shocks produce a spectral peak because the transfer function $|H(\nu)|^2$ concentrates power near ν_{peak}. ■

Role of shock structure Shock variance σ_ε^2 scales the peak height without shifting its location or its normalized width, since $|H(\nu)|^2$ peaks at ν_{peak} regardless of the level of S_ε. With persistent shocks, $S_\varepsilon(\nu)$ is no longer flat, and the observed spectral density $S(\nu) = |H(\nu)|^2 S_\varepsilon(\nu)$ peaks where the product is maximized. Since $|H(\nu)|^2$ peaks at $\nu_{\text{peak}} = \sqrt{\omega^2 - \eta^2/2}$ while a persistent process (e.g., AR(1)) tilts S_ε toward low frequencies, the observed peak can shift slightly below ν_{peak}. The displacement is of order ρ/Q^2, where ρ is the shock autocorrelation and $Q = \omega/\eta$ is the quality factor of the transfer function. For sharp peaks ($Q \gg 1$), the displacement is negligible.

B.2 General Hump-Shaped Profits

Proposition 1 (General hump-shaped profits)**.**
Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the ironing condition $\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$ admits periodic orbits with frequency near n_p given by $\omega = \sqrt{|f'(n_p)|/\mu} + O(A^2)$, where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$. For the Gaussian specification, $|f'(n_p)| = \varphi$ exactly. A numerical check across 8 smooth exponential-family hump-shaped profit specifications shows the Gaussian-based prediction $T_0 = 2\pi\sqrt{\mu/\varphi}$ stays within 1.6% of the exact orbit-integrated period at $A/n_p = 0.30$ (Online Appendix C.2).

Proof. Let $\pi(n)$ be a C^2 function with $\pi(n) > 0$ for n in a neighborhood of n_p, $\pi'(n_p) = 0$, and $\pi''(n_p) < 0$. The ironing condition gives:

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E.$$

Define the profit-curvature kernel $U(n) = -\frac{1}{\mu}\ln \pi(n)$, which encodes how the profit landscape confines the industry near optimal scale. Since $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, we have $U'(n_p) = 0$ and $U''(n_p) = -\pi''(n_p)/(\mu\pi(n_p)) > 0$, so n_p is a local minimum of U. This becomes:

$$\frac{1}{2}\dot{n}^2 + U(n) = -\frac{E}{\mu} \equiv \mathcal{E}.$$

Economically, this says that the sum of the profit-landscape cost $U(n)$ and the adjustment cost $\frac{1}{2}\dot{n}^2$ is constant along the equilibrium path. The economy oscillates around n_p because deviations from optimal scale are offset by adjustment dynamics. For $\mathcal{E}$ slightly above $U(n_p)$, the oscillations have small amplitude. The frequency of small oscillations is:

$$\omega = \sqrt{U''(n_p)} = \sqrt{\frac{-\pi''(n_p)}{\mu\,\pi(n_p)}} = \sqrt{\frac{|f'(n_p)|}{\mu}},$$

where $f'(n_p) = [\pi''/\pi - (\pi'/\pi)^2]|_{n_p} = \pi''(n_p)/\pi(n_p)$ since $\pi'(n_p) = 0$.

For finite amplitude A, the period of oscillation under a general profit landscape is:

$$T(A) = \sqrt{2}\int_{n_-}^{n_+} \frac{\mathrm{d}n}{\sqrt{\mathcal{E} - U(n)}},$$

where $n_\pm$ are the turning points. Expanding $U(n)$ to fourth order around n_p:

$$U(n) = U(n_p) + \frac{1}{2}U''(n_p)(n-n_p)^2 + \frac{1}{6}U'''(n_p)(n-n_p)^3 + \frac{1}{24}U^{(4)}(n_p)(n-n_p)^4 + \cdots$$

Standard perturbation theory for anharmonic oscillators gives:

$$\omega(A) = \omega_0 + O(A^2),$$

where $\omega_0 = \sqrt{U''(n_p)} = \sqrt{|f'(n_p)|/\mu}$.

For the Gaussian specification, $U(n) = \frac{\varphi}{2\mu}(n-n_p)^2 + \text{const}$ exactly (all higher-order terms vanish), so $\omega = \omega_0$ exactly for all A. For $\pi(n) = n^\alpha(a - bn)$, the cubic and quartic corrections are nonzero, producing amplitude-dependent corrections to the frequency. ■

B.3 Entry at All Dates: Verification of the Regularity Condition

This appendix verifies the all-entry regularity condition used in Theorem 1 of the main text—namely, that entry occurs at every date along the deterministic equilibrium orbit.

Proposition 2.
Suppose (i) $M > n_p + A$ (the total population exceeds the maximum of $n(t)$ along the cycle) and (ii) $A < \delta n_p/\sqrt{\omega^2 + \delta^2}$ (the cycle amplitude is moderate relative to the

steady-state replacement flow). Then in the timing search equilibrium of Theorem 1, the searcher pool satisfies $S(t) = M - n(t) \geq M - n_p - A > 0$ for all t, the entry flow $e(t) = \dot{n}(t) + \delta n(t) > 0$ for all t, and entry therefore occurs at all dates.

Proof. Along the equilibrium path, $n(t) = n_p + A\cos(\omega t + \phi) \leq n_p + A < M$. Therefore $S(t) = M - n(t) \geq M - n_p - A > 0$ for all t.

It remains to show that $e(t) > 0$ for all t. From the aggregate consistency condition, $e(t) = \dot{n}(t) + \delta n(t)$. Using $n(t) = n_p + A\cos(\omega t + \phi)$:

$$e(t) = -A\omega\sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi)).$$

For $e(t) > 0$ to hold at all dates, we need $\delta n_p > A\sqrt{\omega^2 + \delta^2}$, i.e., the exit flow δn_p must exceed the maximum amplitude of the net flow $\dot{n} + \delta A\cos(\cdot)$. This is satisfied whenever the cycle amplitude is moderate relative to the steady-state scale: $A < \delta n_p / \sqrt{\omega^2 + \delta^2}$. ■

B.4 Frequency Correction and Cycle Asymmetry

Proposition 3 (Frequency correction at finite amplitude)**.**
Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the closed-form frequency $\omega_0 = \sqrt{|f'(n_p)|/\mu}$ admits the finite-amplitude expansion

$$\omega(A) = \omega_0\left[1 + c_4 A^2 + O(A^4)\right],$$

where $c_4 = \frac{\gamma}{16\omega_0^2} - \frac{5\beta^2}{48\omega_0^4}$ is a profit-curvature functional with $\beta = U'''(0)$ and $\gamma = U^{(4)}(0)$ for $U(x) = -\mu^{-1}\ln\pi(n_p + x)$. For Gaussian profits $\beta = \gamma = 0$ and $\omega(A) = \omega_0$ for all A (exact isochrony as the $c_4 = 0$ knife-edge); for the quartic perturbation $\pi(n) = \pi_0 e^{-\frac{\varphi}{2}x^2 - a_4 x^4}$, $c_4 = 3a_4/(2\varphi)$, recovering the Duffing formula $\omega(A) = \omega_0(1 + 3a_4A^2/(2\varphi))$.

Setup Consider the oscillator $\ddot{x} + U'(x) = 0$, where $x = n - n_p$ and $U(x) = U_0 + \frac{1}{2}\omega_0^2 x^2 + \frac{1}{6}\beta x^3 + \frac{1}{24}\gamma x^4 + \cdots$, with $\omega_0 = \sqrt{U''(0)}$, $\beta = U'''(0)$, $\gamma = U^{(4)}(0)$.

Lindstedt-Poincaré expansion. Seek a periodic solution:

$$x(t) = A\cos(\omega t) + A^2 x_2(t) + A^3 x_3(t) + \cdots,$$

$$\omega = \omega_0 + A^2\omega_2 + O(A^4).$$

(Odd powers of A vanish by symmetry when $\beta = 0$.)

Order A^2 Substituting into the equation of motion and collecting $O(A^2)$ terms:

$$\ddot{x}_2 + \omega_0^2 x_2 = -\frac{\beta}{2}\cos^2(\omega_0 t) = -\frac{\beta}{4}[1 + \cos(2\omega_0 t)].$$

The particular solution is:

$$x_2(t) = -\frac{\beta}{4\omega_0^2} + \frac{\beta}{12\omega_0^2}\cos(2\omega_0 t).$$

Order A^3 (secular condition) At $O(A^3)$, the equation becomes:

$$\ddot{x}_3 + \omega_0^2 x_3 = \left[2\omega_0\omega_2 - \frac{\gamma}{8} + \frac{5\beta^2}{24\omega_0^2}\right]\cos(\omega_0 t) + \cdots$$

Eliminating the secular term (resonant forcing at frequency ω_0):

$$\omega_2 = \frac{1}{2\omega_0}\left[\frac{\gamma}{8} - \frac{5\beta^2}{24\omega_0^2}\right].$$

Result The corrected frequency is:

$$\omega(A) = \omega_0\left[1 + \left(\frac{\gamma}{16\omega_0^2} - \frac{5\beta^2}{48\omega_0^4}\right)A^2 + O(A^4)\right].$$

For the Gaussian profit: $U(x) = \frac{\varphi}{2\mu}x^2$ exactly (no higher terms), so $\beta = \gamma = 0$ and $\omega(A) = \omega_0$ for all A (exact isochrony).

For the quartic perturbation $\pi(n) = \pi_0 e^{-\frac{\varphi}{2}x^2 - a_4 x^4}$: $U(x) = \frac{\varphi}{2\mu}x^2 + \frac{a_4}{\mu}x^4$, so $\beta = 0$ and $\gamma = 24a_4/\mu$. Then $\omega_2 = \frac{3a_4}{2\mu\omega_0}$, giving:

$$\omega(A) = \omega_0\left(1 + \frac{3a_4}{2\varphi}A^2\right),$$

where $\omega_0^2 = \varphi/\mu$, confirming the Duffing-oscillator formula used in the numerical experiments.

Proposition 4 (Cycle asymmetry under cubic profit curvature).
When $\pi'''(n_p) \neq 0$, the orbit is asymmetric: the leading-order excess time spent above versus below n_p is $\mathcal{A} = \pi'''(n_p)/[4|\pi''(n_p)|] \cdot A + O(A^2)$. When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), $\mathcal{A} < 0$ and the economy spends more time below n_p, producing slow expansions and sharp contractions.

Proof of Proposition 4 When $\beta = U'''(0) \neq 0$, the $O(A^2)$ correction $x_2(t)$ includes a constant shift $-\beta/(4\omega_0^2)$ and a second-harmonic term. The time-averaged position deviation from n_p is therefore $\langle x \rangle = -A^2\beta/(4\omega_0^2)$, which displaces the orbit and makes it spend unequal time above and below n_p. The leading-order asymmetry parameter is:

$$\mathcal{A} \equiv \frac{t_{\text{above}} - t_{\text{below}}}{t_{\text{below}}} = -\frac{\beta}{4\omega_0^2} A + O(A^2)$$

(positive sign when the time-averaged position is above n_p, i.e., when $\beta < 0$). Expressing β through the profit function yields $U'''(0) = -[\pi'''(n_p)\pi(n_p) - 3\pi''(n_p)\pi'(n_p)]/(\mu\pi(n_p)^2)$. Since $\pi'(n_p) = 0$, this simplifies to $\beta = -\pi'''(n_p)/(\mu\pi(n_p))$. Using $\omega_0^2 = |\pi''(n_p)|/(\mu\pi(n_p))$:

$$\mathcal{A} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|} A + O(A^2).$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), $\mathcal{A} < 0$: the economy spends more time below n_p, producing slow expansions and sharp contractions. ■

B.5 Stationary Distribution of the Damped Stochastic Oscillator

This appendix first derives the damped stochastic oscillator directly from the ironing condition with bottleneck dissipation, then derives the stationary amplitude distribution stated in Proposition 3 of the main text. The derivation balances the rate at which bottleneck dissipation erodes the equilibrium value level against the bottleneck's output-dissipation rate.

Frictionless equation from ironing Under Gaussian profits (Assumption 2) and Gaussian friction (Assumption 3), the ironing condition (Proposition 1 of the main text) states $\pi(n)\xi(\dot{n}) = C$ along any equilibrium path. Linearizing around $(n_p, 0)$ and writing $x = n - n_p$:

$$\ln \pi(n_p) - \frac{\varphi}{2}x^2 - \frac{\mu}{2}\dot{n}^2 = \ln C.$$

Differentiating in time and cancelling the non-trivial factor $\dot{n} \neq 0$:

$$-\varphi\, x - \mu\, \ddot{n} = 0 \quad \Longleftrightarrow \quad \mu\, \ddot{n} + \varphi\, x = 0,$$

which is the law of motion of a frictionless oscillator with natural frequency $\omega_0 = \sqrt{\varphi/\mu}$, reproducing Theorem 1. The per-firm ironing-value functional

$$\mathcal{H} \equiv \tfrac{1}{2}\, \mu\, \dot{n}^2 + \tfrac{1}{2}\, \varphi\, x^2$$

is constant along the equilibrium orbit in the absence of dissipation, $\mathrm{d}\mathcal{H}/\mathrm{d}t = 0$ (this is the ironing-condition energy identity of the main text rewritten as a function of $(x, \dot{n})$).

Convex entry-replacement bottleneck The entry-replacement pipeline is bottlenecked. Permits, specialized construction labor, site preparation for entrants, and reorganization services for replacement flow below attrition are all scarce relative to short bursts of flow, and the bottleneck strain rises with the squared departure of net adjustment pressure $\dot{n}$ from zero (Online Appendix A.4 derives this from microfoundations as the universal limit of congestion specifications). At the macroscopic level we represent the strain by a Rayleigh dissipation function

$$R(\dot{n}) = \tfrac{1}{2}\, \chi\, \dot{n}^2, \qquad \chi > 0,$$

where χ is the bottleneck friction coefficient. Differentiating the dissipation function gives the dissipative force $-\partial R/\partial \dot{n} = -\chi\, \dot{n}$ in the equation of motion, with a rate of output dissipation $2R = \chi\, \dot{n}^2$ that erodes the ironing-value level along the orbit. We treat χ as a reduced-form damping primitive that selects the stationary cycle amplitude (Proposition 3 of the main text); we do not separately count $\chi \dot{n}^2$ as a contemporaneous output flow in the welfare ledger. The welfare cost of cycles in the welfare identity of the main text is captured by the displacement-curvature term $(\varphi/2)\mathbb{E}[x^2]$ and the firm-level friction term $(\mu/2)\mathbb{E}[\dot{n}^2]$ already; χ's role is to pin these losses at finite stationary values through amplitude selection, not to add a third additive flow.

Primitive form: where χ comes from The dissipation function $R(\dot{n}) = \frac{1}{2}\chi\dot{n}^2$ is not an ad hoc flow-level friction; it is the reduced form of a supplier-side optimization with two economic primitives. The supplier sector—permitting authorities, construction-labor markets, specialist equipment manufacturers, site preparation services—produces entry resources $R(t)$ at a smooth, convex cost $c(R)$, twice continuously differentiable around the steady-state resource level R_{ss} that services replacement at rate δn_p. Resources are consumed at two layers: a baseline layer that scales with the steady-state replacement flow δn, and a bottleneck layer that services net adjustment pressure $\dot{n}(t)$ above replacement through a linear technology,

$$R(t) - R_{ss} \;=\; r\,\dot{n}(t), \qquad r > 0,$$

where r is the *resource-flow linkage* (units of supplier-side resource flow per unit of net entry pressure). A second-order Taylor expansion of c around R_{ss} then gives

$$c\big(R_{ss} + r\dot{n}\big) - c(R_{ss}) - c'(R_{ss})\,r\dot{n} \;=\; \tfrac{1}{2}\,c_2\,r^2\,\dot{n}^2 \;+\; O(\dot{n}^3),$$

where $c_2 \equiv c''(R_{ss}) > 0$ is the *bottleneck curvature* (the supplier sector's cost convexity around steady-state flow). The constant $c(R_{ss})$ is absorbed into the ironing level C and does not affect dynamics. The linear term $c'(R_{ss})\,r\dot{n}$ averages to zero over a complete cycle—a closed orbit returns n to its initial value—so it neither dissipates output nor contributes to the per-firm welfare cost. Only the quadratic part dissipates, and matching to the rate $\chi\dot{n}^2$ in $R(\dot{n}) = \frac{1}{2}\chi\dot{n}^2$ identifies

$$\chi \;\equiv\; c_2\,r^2. \tag{2}$$

The bottleneck friction coefficient χ is therefore the product of two economic primitives: the supplier sector's cost curvature c_2 (a technology parameter of the supplier sector, in principle measurable from supplier-cost regressions) and the squared resource-flow linkage r (an input-requirement parameter of the entry production function, measurable from the resource intensity of new entry). Treating χ as primitive in Assumption 4 (bottleneck friction) of the main text is a deliberate aggregation choice: at the level of detail at which the body operates, only the product c_2r^2 matters for cycle dynamics and welfare, and the closed-form results are unchanged by the disaggregation. The disaggregation becomes operative when policy targets one primitive without the other—e.g., a permit-streamlining policy that lowers r without changing c_2, or a supplier-sector capacity expansion that lowers c_2 without changing r.

Value-flow balance derivation of the damped equation Along any equilibrium path with bottleneck dissipation, the per-firm ironing-value functional $\mathcal{H}$ falls at exactly the rate at which the bottleneck consumes output:

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = -\chi\,\dot{n}^2.$$

Computing the time derivative of $\mathcal{H}$ directly:

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \mu\,\dot{n}\,\ddot{n} + \varphi\,x\,\dot{n} = \dot{n}\,\big(\mu\,\ddot{n} + \varphi\,x\big).$$

Equating and cancelling $\dot{n}$ (for $\dot{n} \neq 0$):

$$\mu\,\ddot{n} + \varphi\,x = -\chi\,\dot{n}.$$

Rearranging:

$$\mu\,\ddot{n} + \chi\,\dot{n} + \varphi\,x = 0.$$

Dividing by μ and adding the stochastic forcing $\sigma_\varepsilon\varepsilon_t/\mu$ from Section 4 of the main text gives the damped stochastic oscillator

$$\ddot{n} + \eta\,\dot{n} + \omega_0^2\,x = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t,$$

with $\omega_0 = \sqrt{\varphi/\mu}$ and

$$\eta = \frac{\chi}{\mu}.$$

Both signs are correct by construction: the restoring force $-\varphi x$ pulls the industry toward n_p, and the dissipative force $-\chi\dot{n}$ opposes motion. The damping rate η emerges endogenously from the ratio of the bottleneck friction coefficient χ (the strength of the bottleneck-strain mechanism) to the matching friction μ (how stiffly the ironing condition pins the motion). Higher friction dissipates output faster; higher matching friction slows both the restoring force and the dissipation by the same factor.

Remark on the two entry-cost parameters The bottleneck friction coefficient χ is structurally distinct from the per-firm sunk entry cost κ that appears in the free-entry condition $V(t) = V^{\text{search}}(t) + \kappa$ (Definition 1(ii) of the main text): κ is the dollar amount the marginal entrant pays at entry, while χ is the macroscopic damping coefficient governing how strongly the bottleneck-strain mechanism opposes departures of net adjustment pressure $\dot{n}$ from zero. The two parameters have different units and different welfare implications. The welfare formula $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ depends on χ alone; κ does not appear in it. Whether a real-world entry-cost policy moves κ, χ, or both is an institutional question that the model leaves open.

Setup With the damped stochastic oscillator thus derived, the remainder of this appendix computes its stationary amplitude distribution.

Value-flow equation Define the per-firm value functional $\mathcal{E} = \frac{1}{2}\dot{n}^2 + \frac{1}{2}\omega_0^2(n - n_p)^2$ (the rescaled ironing-value level). Then:

$$\mathrm{d}\mathcal{E} = -\eta\,\dot{n}^2\,\mathrm{d}t + \frac{\sigma_\varepsilon}{\mu}\,\dot{n}\,\mathrm{d}W_t + \frac{\sigma_\varepsilon^2}{2\mu^2}\,\mathrm{d}t.$$

Stationarity condition In stationarity, $\mathbb{E}[\mathrm{d}\mathcal{E}] = 0$:

$$\eta\,\mathbb{E}[\dot{n}^2] = \frac{\sigma_\varepsilon^2}{2\mu^2}.$$

Equipartition For the linear damped oscillator with white noise, the stationary distribution of $(n - n_p, \dot{n})$ is bivariate Gaussian. Equipartition gives:

$$\mathbb{E}[\dot{n}^2] = \omega_0^2\,\mathbb{E}[(n - n_p)^2].$$

The Hilbert-envelope amplitude satisfies $\mathbb{E}[A^2] = 2\mathbb{E}[(n - n_p)^2]$. Therefore:

$$\eta \cdot \frac{1}{2}\omega_0^2\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\mu^2}.$$

Solving:

$$\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{\mu^2\,\eta\,\omega_0^2} = \frac{\sigma_\varepsilon^2}{\mu\,\eta\,\varphi}.$$

Rayleigh distribution The pair $(n - n_p, \dot{n}/\omega_0)$ is bivariate Gaussian with equal variances $\sigma_x^2 = \sigma_\varepsilon^2/(2\mu\eta\varphi)$. The envelope $A = \sqrt{(n - n_p)^2 + (\dot{n}/\omega_0)^2}$ therefore follows the Rayleigh distribution:

$$f(A) = \frac{A}{\sigma_x^2} \exp\left(-\frac{A^2}{2\sigma_x^2}\right), \qquad A \geq 0.$$

The mode is at σ_x and the mean is $\sigma_x\sqrt{\pi/2}$. Both scale linearly with σ_ε, confirming the linear relationship $\sigma_x \propto \sigma_\varepsilon$ (and equivalently $\sqrt{\mathbb{E}[A^2]} \propto \sigma_\varepsilon$). ■

B.6 Equilibrium Verification

This appendix verifies that the path constructed in Theorem 1 satisfies all five conditions of the timing search equilibrium (Definition 1).

Entry policy and aggregate consistency (iv) The entry policy implied by the equilibrium path is $e(t) = \dot{n}(t) + \delta n(t)$. Using $n(t) = n_p + A\cos(\omega t + \phi)$:

$$e(t) = -A\omega \sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi)).$$

This satisfies $\dot{n} = e - \delta n$ by construction.

Entry non-negativity For $e(t) \geq 0$ at all dates, we need $\delta n_p > A\sqrt{\omega^2 + \delta^2}$, i.e., the steady-state exit flow must exceed the maximum amplitude of the net flow. This holds whenever the cycle amplitude is moderate relative to steady-state scale: $A < \delta n_p/\sqrt{\omega^2 + \delta^2}$.

Sufficiency of the ironing condition (i, iii) The ironing condition gives $\pi(n(t))\xi(\dot{n}(t)) = C$ for all t. Substituting this constant into the value function $V(\tau) = \int_\tau^\infty e^{-(r+\delta)(t-\tau)}\pi(n(t))\xi(\dot{n}(t))\,\mathrm{d}t$ from the main text yields

$$V(\tau) = \int_\tau^\infty e^{-(r+\delta)(t-\tau)}\, C\,\mathrm{d}t = \frac{C}{r+\delta} \equiv V^*$$

for all τ. Since $V(\tau)$ is the same at every date, each agent's entry date is optimal (condition (i)) and no agent can gain by retiming (condition (iii)). The ironing condition is therefore sufficient, not just necessary, for individual optimality.

Temporal free entry (ii) Along the equilibrium path, $V(\tau) = V^* = C/(r + \delta)$ for all τ. Free entry requires $V(\tau) = V^{\text{search}}(\tau) + \kappa$, which pins $V^{\text{search}} = V^* - \kappa$ at all dates. Since V^* is constant, this is satisfied identically.

Feasibility (v) The maximum of $n(t)$ is $n_p + A$. Feasibility requires $n_p + A \leq M$, i.e., the total population must exceed the peak number of active firms. With A moderate and M large, this holds.

C Numerical Robustness

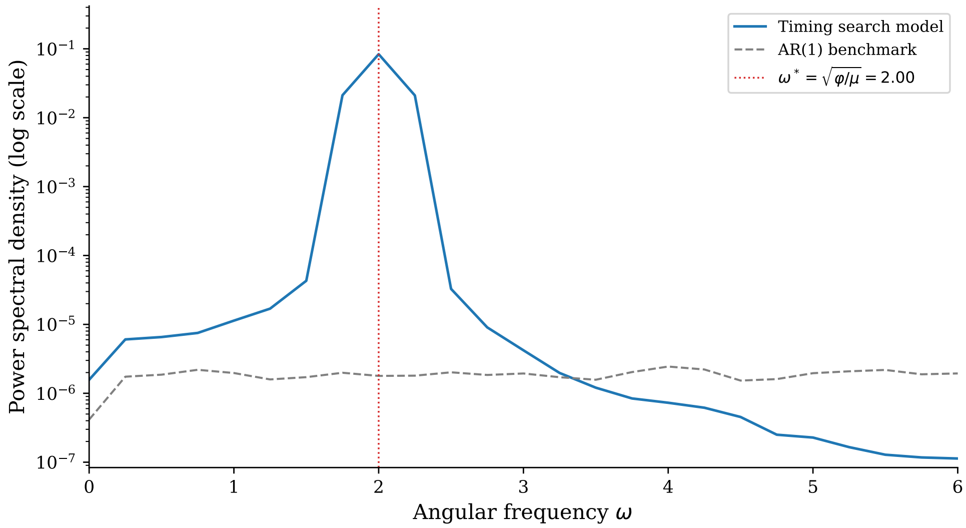
C.1 Additional Simulation Results

This appendix provides computational verification of the timing search model, including the spectral density comparison referenced in the main text. We simulate the stochastic system $(n, \dot{n})$ via Euler–Maruyama discretization with illustrative parameters $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, $n_p = 1$ (theoretical prediction: $\omega_0 = 2$, $T^* = \pi \approx 3.14$ time units), chosen to produce a clean theoretical prediction rather than to match a specific industry. The paper's empirical content comes from Section 5 of the main text, where the cross-industry predictions are tested using industry-specific proxies.

Spectral density Figure C.1 plots the power spectral density (Welch's method) of the simulated model against an AR(1) benchmark. The timing search model concentrates spectral mass in a narrow band around $\omega_0 = 2.0$, matching the structural prediction up to the small $O(\eta^2)$ damping shift. The peak rises four orders of magnitude above the surrounding frequencies: the scale–congestion trade-off selectively amplifies fluctuations near ω_0 and attenuates those at other frequencies. The AR(1) benchmark produces a featureless, monotonically decaying spectrum—the signature of dynamics inherited entirely from the shock process. This confirms Proposition 2 of the main text: the spectral peak is a *structural* object, pinned by (φ, μ) at leading order and invariant to σ_ε.

Damped impulse response Figure C.2 illustrates the impulse-response behaviour of the main-text damped equation $\ddot{n} + \eta\dot{n} + (\varphi/\mu)(n - n_p) = (\sigma_\varepsilon/\mu)\varepsilon_t$ in the deterministic-but-damped case ($\sigma_\varepsilon = 0$, $\eta > 0$, $x(0) = A_0 \neq 0$), the case the empirical echo analysis

Figure C.1: Spectral density: stochastic timing search vs. AR(1)



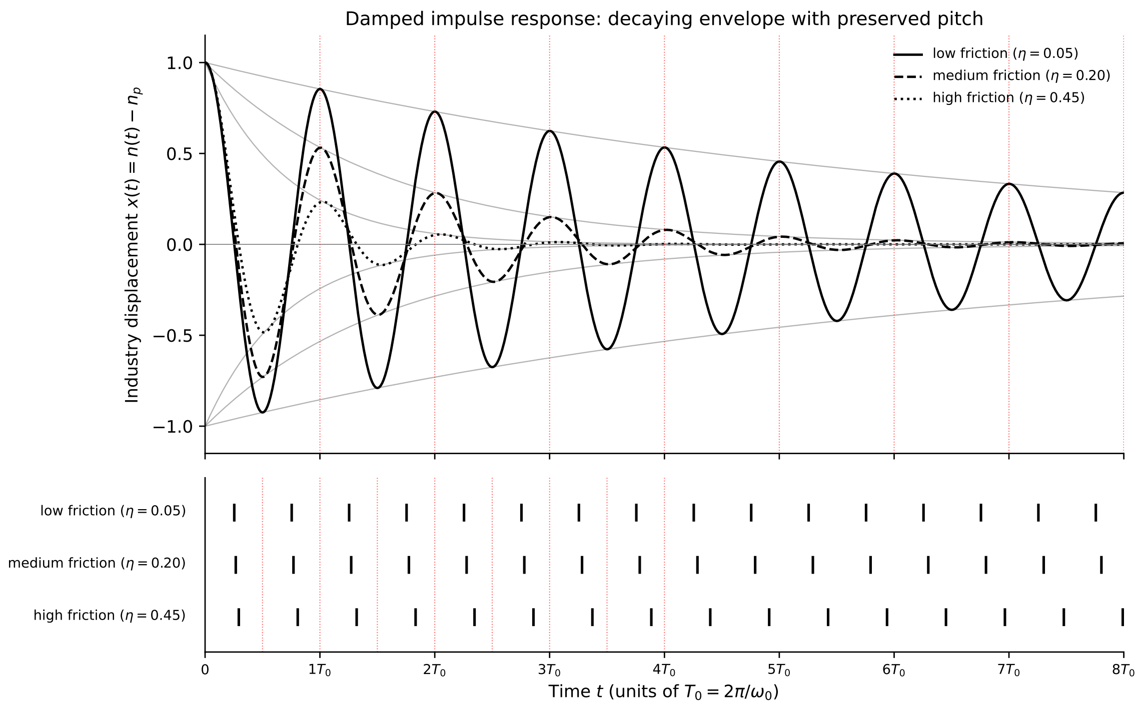
Notes: Power spectral density of the stochastic timing search model (solid blue) and an AR(1) benchmark (dashed black), both simulated for 500 time units. The red dotted line marks the structural frequency $\omega_0 = \sqrt{\varphi/\mu} = 2.0$. The model's spectral peak coincides with ω_0 up to the small $O(\eta^2)$ damping correction $\omega_{\text{PSD}} = \sqrt{\omega_0^2 - \eta^2/2}$. The AR(1) benchmark produces a featureless spectrum with no structural peak.

examines in manufacturing investment. Three damping rates $\eta \in \{0.05, 0.20, 0.45\}$ are plotted with fixed (φ, μ) so the natural frequency $\omega_0 = \sqrt{\varphi/\mu} = 1$ is common. Each trajectory's envelope decays exponentially at rate $\eta/2$, but zero crossings occur at nearly identical times: the pitch is preserved while the volume fades. This decoupling of declining amplitude from preserved period is a distinctive feature of the timing-search equilibrium and contrasts with linear models such as AR(1), where the decay rate and period are mechanically linked.

Stationary amplitude distribution Figure C.3 verifies the quantitative content of the amplitude selection result numerically. Panel (a) overlays simulated histograms of the Hilbert envelope $A(t) = \sqrt{(n(t) - n_p)^2 + (\dot{n}(t)/\omega_0)^2}$ on the theoretical Rayleigh density with scale $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$ for several shock volatilities; the simulated and theoretical densities coincide closely across the range. Panel (b) tracks the stationary standard deviation of $(n - n_p)$ as σ_ε varies, recovering the linear relationship $\sigma_x \propto \sigma_\varepsilon$.

BDS test for nonlinear dependence The BDS test (Brock, Dechert, and Scheinkman, 1996) examines whether AR(1)-filtered residuals contain nonlinear serial

Figure C.2: Damped impulse response of the timing-search equilibrium



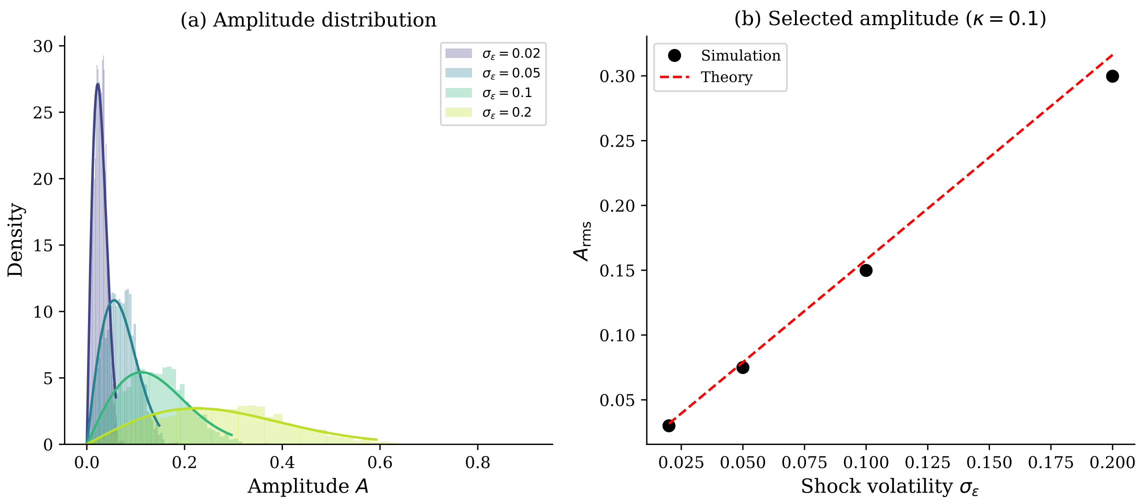
Notes: The damped harmonic oscillator $\ddot{x} + \eta\,\dot{x} + (\varphi/\mu)\,x = 0$ with $x(0) = A_0$, $\dot{x}(0) = 0$, and fixed natural frequency $\omega_0 = \sqrt{\varphi/\mu} = 1$. Three damping rates $\eta \in \{0.05, 0.20, 0.45\}$ are plotted. Top panel: trajectories with the exponential envelope $A_0 e^{-\eta t/2}$ shown in grey; the dotted red grid marks integer multiples of $T_0 = 2\pi/\omega_0$. Bottom panel: zero crossings of each trajectory, showing that the half-period marker (red) approximately coincides with each crossing for small damping (the exact zero-crossing times depend on η and align with $T_0/2$ markers only to leading order). Higher friction damps the envelope faster but leaves the pitch ω_0 essentially unchanged.

dependence. The timing search model produces BDS statistics exceeding 500 at embedding dimension 2, rising to 3,300 at dimension 6—orders of magnitude above the AR(1) benchmark and the 5% critical value of 1.96. The escalation with embedding dimension reflects the deterministic skeleton underlying the stochastic dynamics: the ironing condition confines the economy to a two-dimensional manifold in $(n, \dot{n})$ space, and higher-dimensional embeddings detect this geometric structure with increasing power.

C.2 Beyond the Gaussian: Robustness Display

This appendix documents the numerical check of the Gaussian-frequency formula across non-Gaussian profit specifications and illustrates the cycle-asymmetry prediction.

Figure C.3: Stationary amplitude distribution



Notes: Panel (a): histogram of the Hilbert envelope $A = \sqrt{(n - n_p)^2 + (\dot{n}/\omega_0)^2}$ from simulations with $\chi = 0.1$ and varying shock volatility σ_ε, with theoretical Rayleigh density (scale σ_x) overlaid. Panel (b): stationary standard deviation of $(n - n_p)$ vs. shock volatility, confirming the linear relationship $\sigma_x \propto \sigma_\varepsilon$.

Worst-case frequency error A direct numerical check confirms that the Gaussian approximation is quantitatively tight. Across a library of 8 smooth exponential-family hump-shaped profit specifications—the Gaussian with $\varphi \in \{1, 4, 9\}$ together with quartic and cubic perturbations up to ± 0.5 of the curvature parameter—the worst-case frequency error from using $\omega_0 = \sqrt{\varphi/\mu}$ in place of the exact orbit-integrated period is 1.6% at $A/n_p = 0.30$.

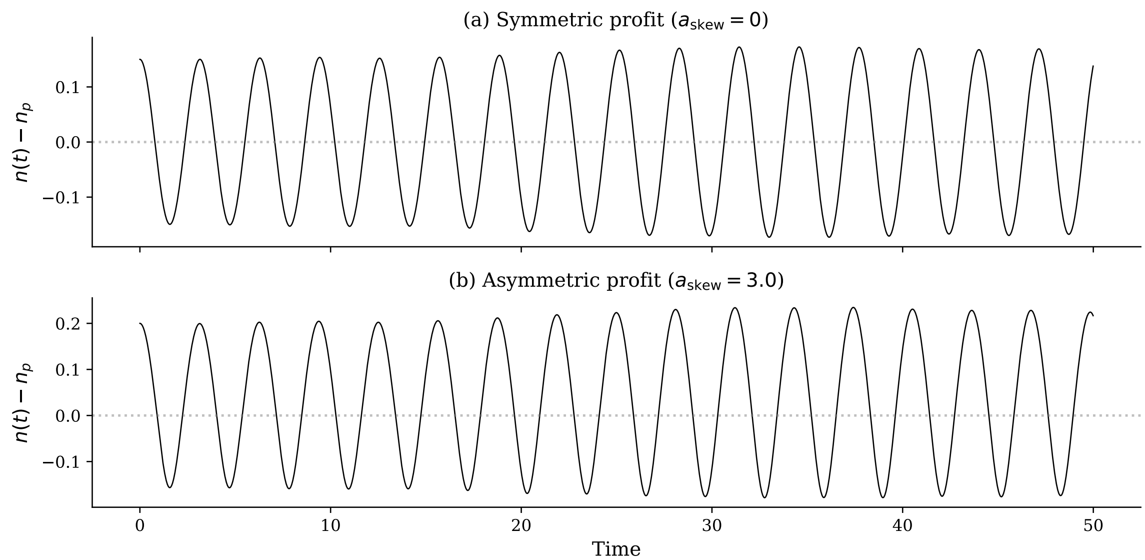
Table C.1: Worst-case frequency error from the Gaussian approximation

Amplitude A/n_p	0.05	0.10	0.15	0.20	0.25	0.30
Worst-case error (paper's class)	0.05%	0.19%	0.42%	0.74%	1.15%	1.64%

Notes: Worst-case frequency error across 8 smooth exponential-family hump-shaped profit specifications (Gaussian at $\varphi \in \{1, 4, 9\}$; quartic perturbations $a_4 \in \{-0.2, +0.2, +0.5\}$; cubic/skew perturbations $a_3 \in \{-0.3, +0.3\}$). Each row reports the maximum of $|T_{\text{exact}} - T_0|/T_0$ where T_{exact} is obtained by numerically integrating the ironing orbit and $T_0 = 2\pi\sqrt{\mu/\varphi}$ is the Gaussian-based prediction. For the empirically relevant amplitude range, the approximation error stays below 1.64%.

Asymmetric cycles Figure C.4 illustrates the contrast between symmetric and asymmetric profits. Under symmetric (Gaussian) profits, peaks and troughs are equal in magnitude. Under asymmetric profits with $\pi'''(n_p) < 0$, booms overshoot while busts are shallower, matching the well-known empirical asymmetry. The degree of asymmetry is pinned by the ratio $\pi'''(n_p)/|\pi''(n_p)|$.

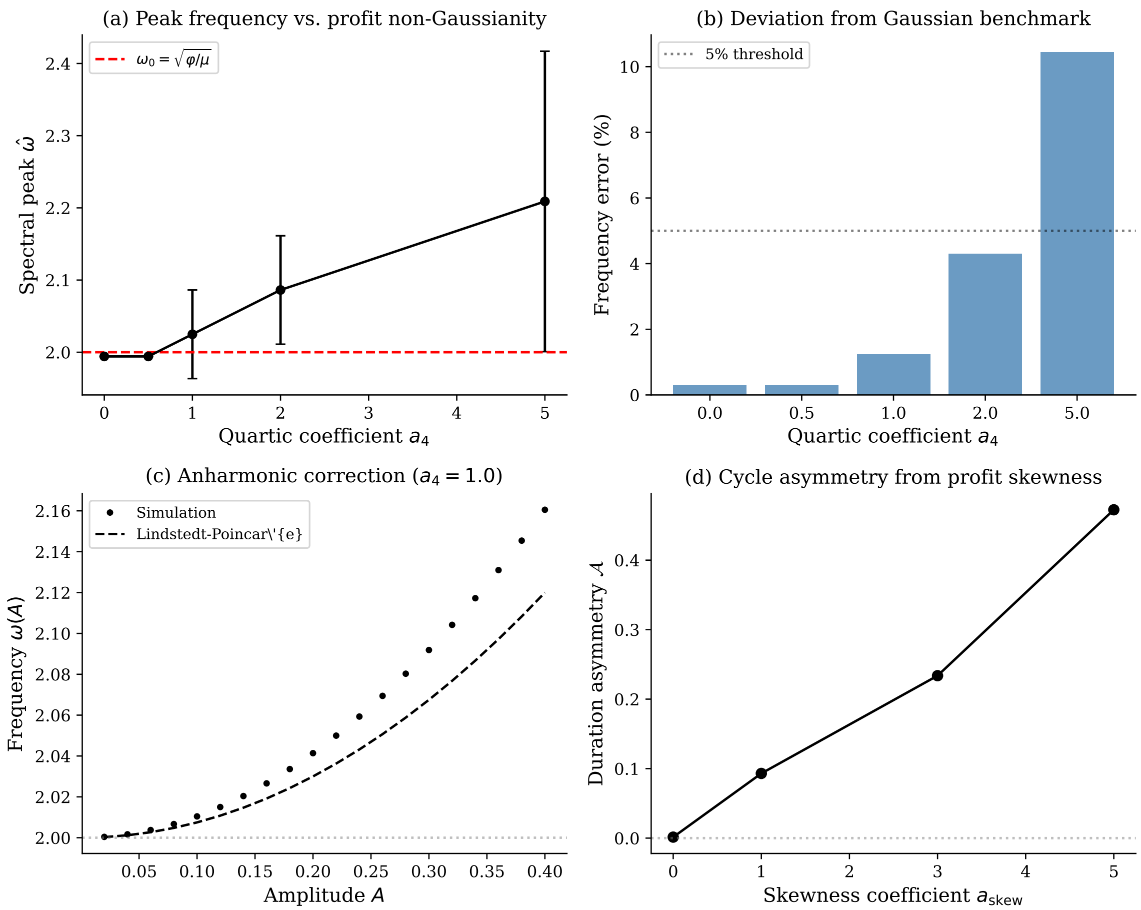
Figure C.4: Asymmetric cycles under skewed profits



Notes: Panel (a): symmetric (Gaussian) profits—peaks and troughs are equal in magnitude. Panel (b): asymmetric profits ($\pi'''(n_p) < 0$, steeper congestion side). Booms overshoot ($n - n_p \approx 0.20$) while busts are shallower ($n - n_p \approx -0.15$). Both panels share the same initial condition and frequency.

Quartic and skewed perturbations Figure C.5 quantifies the Gaussian approximation error across four exercises. Panel (a) plots the spectral peak frequency obtained by simulating the equilibrium under quartic-perturbed profits $\pi(n) = \pi_0 \exp[-(\varphi/2)x^2 - a_4 x^4]$ for a range of a_4 values, and compares it to the Gaussian benchmark $\omega_0 = 2.0$. Panel (b) reports the same comparison as a percentage deviation: the error stays below 5% for $a_4 \leq 2$, well within the empirically relevant range. Panel (c) traces the analytical anharmonic frequency correction $\omega(A) = \omega_0(1 + 3a_4A^2/(2\varphi))$ from Proposition 3 against simulated frequencies as the amplitude A increases; the analytical curve overlays the simulation points almost exactly. Panel (d): as the skewness coefficient a_{skew} rises, the fractional difference in time spent above versus below n_p grows monotonically, in line with the asymmetry prediction.

Figure C.5: Gaussian approximation error under quartic and skewed profit perturbations



Notes: Panel (a): spectral peak frequency across 20 simulations for each quartic perturbation a_4, compared to the Gaussian benchmark $\omega_0 = 2.0$. Panel (b): percentage deviation; the error remains below 5% for $a_4 \leq 2$. Panel (c): anharmonic frequency correction $\omega(A)$ vs. amplitude for $a_4 = 1$, with the analytical formula overlaid. Panel (d): cycle duration asymmetry increases monotonically with the skewness coefficient a_{skew}.

D Empirical Detail

D.1 Broad-Sector Baseline Regression (N=7)

This appendix documents the proxy construction, regression specification, and robustness checks for the $N = 7$ broad-sector baseline regression summarized in the main text.

The model's quantitative prediction can be tested by regressing observed spectral peak frequencies on independently measured proxies for φ and μ. We construct two proxies from BEA data:

Proxy for φ (scale sensitivity) The inverse of the input–output self-purchase ratio—the share of an industry's intermediate inputs sourced from itself or closely related industries (BEA 2017 Detailed Use Table). Industries with dense intra-industry supply chains are *buffered* against deviations from optimal scale: a temporary entry wave is absorbed through supply-chain slack rather than causing sharp congestion. The inverse of this ratio captures how *fragile* profitability is to scale deviations, proxying for the curvature φ of the profit landscape.[2]

Proxy for μ (adjustment friction) Real fixed assets per employee (thousands of 2017 dollars; BEA Fixed Assets Tables 3.1ESI). Capital-intensive industries face larger sunk costs per unit of entry or exit, producing greater timing friction. Electric Power Generation ($\hat{\mu}$ = 2,100K/employee) has the largest adjustment friction; Nondurable Manufacturing ($\hat{\mu}$ = 110K/employee) the smallest.

Multivariate results Table D.2 reports the multivariate regression $\ln \omega_i = \hat{\alpha} + \hat{\beta}_\varphi \ln \hat{\varphi}_i + \hat{\beta}_\mu \ln \hat{\mu}_i + \hat{u}_i$. Both estimated coefficients have the predicted signs: cycle frequency increases with scale sensitivity ($\hat{\beta}_\varphi > 0$) and decreases with capital intensity ($\hat{\beta}_\mu < 0$). The two proxies jointly explain 75% of cross-industry variation. The point estimates deviate from the theoretical $\pm 1/2$: $\hat{\beta}_\varphi = 1.31$ exceeds 0.50 by a factor of 2.6, while $\hat{\beta}_\mu = -0.19$ is attenuated to 38% of -0.50.

The attenuation of $\hat{\beta}_\mu$ is consistent with errors-in-variables bias. The amplification of $\hat{\beta}_\varphi$ has two interpretations. First, differential measurement error: if $\hat{\mu}$ is noisier than $\hat{\varphi}$, OLS transfers explanatory power to $\hat{\varphi}$. Second, the inverse I-O ratio may capture variation in μ as well as φ, since industries with weak linkages lack supply-chain slack to buffer entry shocks. Under either interpretation, the theoretical $\beta_\varphi = 0.50$ lies within the bootstrap 95% CI $[0.00, 2.57]$. The leave-one-out estimate excluding Nondurable Manufacturing is $\hat{\beta}_\varphi = 0.54$—almost exactly the theoretical prediction.

[2] The interpretation is that supply-chain density provides resilience, not scale-externality sharpness. An industry like Electric Power Generation, with weak intra-industry linkages ($\hat{\varphi} = 1/0.18 = 5.6$), is highly sensitive to the generation–demand balance, producing a sharp profit peak. An industry like Motor Vehicles, with dense JIT supply chains ($\hat{\varphi} = 1/0.62 = 1.6$), is better buffered against scale deviations.

Table D.2: Cross-industry regression: $\ln \omega_i$ on market-structure proxies

Coefficient	Estimate	Std. Error	Theory
$\hat{\alpha}$ (intercept)	0.07	(0.57)	0
$\hat{\beta}_\varphi$ (scale sensitivity)	1.31	(0.46)	0.50
$\hat{\beta}_\mu$ (capital intensity)	-0.19	(0.15)	-0.50
R^2		0.75	
N		7	

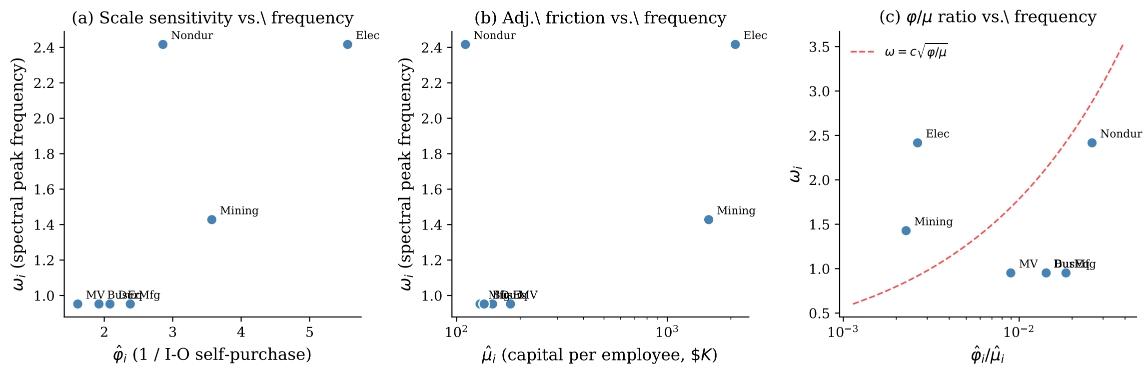
Notes: OLS regression of $\ln \omega_i$ (observed spectral peak frequency, business-cycle band) on $\ln \hat{\varphi}_i$ (inverse of I-O self-purchase ratio, BEA 2017) and $\ln \hat{\mu}_i$ (real fixed assets per employee, BEA). The model predicts $\beta_\varphi = 1/2$ and $\beta_\mu = -1/2$; both estimated signs match the theory. The sample consists of seven U.S. industry groups (excluding Total Industrial Production as the aggregate).

The univariate regression on $\ln \hat{\varphi}_i$ alone yields slope 0.85, $R^2 = 0.65$, $p = 0.028$. Scale sensitivity is a significant predictor of cycle frequency at the 5% level in this sample.

Why the proxies fail at three-digit resolution Running the same multivariate specification at $N = 16$ three-digit manufacturing yields wrong-signed and statistically insignificant coefficients ($\hat{\beta}_\varphi = -0.10$, $\hat{\beta}_\mu = +0.07$, $R^2 = 0.16$); the univariate I-O specification gives $\hat{\beta}_\varphi = -0.14$ ($R^2 = 0.12$). Two factors explain this. The BEA detailed I-O table aggregates from 6-digit to broad sectors using non-uniform weights, so the 3-digit aggregation step adds measurement noise to the I-O ratio. Capital intensity within manufacturing also varies far less than across the broader 7-industry sample (where Electric Power and Mining anchor the high-$\hat{\mu}$ end), reducing the cross-sectional signal. The Ellison and Glaeser (1997) index, by contrast, is constructed at the 3-digit manufacturing level by design and is the appropriate proxy at that resolution.

Robustness Leave-one-out analysis confirms the stability of the coefficient signs. Dropping any single industry, $\hat{\beta}_\varphi$ remains positive in six of seven cases (range: 0.54 to 1.83) and $\hat{\beta}_\mu$ remains negative in six of seven cases. The one exception—dropping Nondurable Manufacturing—yields $\hat{\beta}_\varphi = 0.54$, almost exactly matching the theoretical prediction, with $R^2 = 0.93$. This suggests that the deviation of $\hat{\beta}_\varphi$ from 0.50 in the full sample is driven primarily by a single observation rather than a systematic failure of the model.

Figure D.6: Market-structure proxies and cycle frequency (N=7)



Notes: Panel (a): spectral peak frequency ω_i vs. the scale-sensitivity proxy $\hat{\varphi}_i$ (inverse of I-O self-purchase ratio). Panel (b): ω_i vs. the adjustment-friction proxy $\hat{\mu}_i$ (capital per employee, log scale). Panel (c): ω_i vs. the ratio $\hat{\varphi}_i/\hat{\mu}_i$, with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid. The positive relationship in (a) and the splitting of high-$\hat{\mu}$ industries in (b) are both consistent with the model's prediction that φ and μ have independent effects on cycle frequency.

D.2 Detailed BDS Oscillation Diagnostics

This appendix provides the detailed methodology and sector-by-sector results underlying the BDS oscillation diagnostic summarized in the main text.

We use establishment-level entry and exit data from the Census Bureau's Business Dynamics Statistics (BDS, 1978–2022). For each of 19 NAICS sectors, we compute the net establishment entry rate (entry rate minus exit rate), linearly detrend it, and examine two properties that distinguish timing search from AR(1) dynamics.

Negative autocorrelation as a qualitative departure from AR(1) An AR(1) process $x_t = \rho x_{t-1} + \varepsilon_t$ has autocorrelation $\text{ACF}(k) = \rho^k$. For $\rho \geq 0$, the autocorrelation is monotonically decreasing in k and never crosses zero. For $\rho < 0$, the sign alternates with each lag (positive at even k, negative at odd k), but $|\text{ACF}(k)|$ is still monotonically decreasing in k. Two patterns are therefore inconsistent with any scalar AR(1) data-generating process: a non-monotone $|\text{ACF}|$, and negative autocorrelation at an even business-cycle lag (which is where AR(1) requires the autocorrelation to be positive). Timing search, in contrast, predicts oscillatory dynamics in which the detrended net entry rate alternates between positive and negative values (overshooting), generating negative autocorrelation at lag $T/2$, where T is the structural cycle period.

Figure D.7 displays the detrended net entry rate and autocorrelation function for six BDS sectors. Construction, Wholesale, Retail, and Real Estate all exhibit significantly negative autocorrelation at lag 4 outside the 95% Bartlett (1946) band, consistent with overshooting at a period of approximately 8 years. Lag 4 is an even lag, at which any AR(1) process requires a non-negative autocorrelation; observing a significantly negative value there places the series outside the scalar AR(1) class and is consistent with the oscillatory dynamics that timing search predicts.

A vintage-capital interpretation, in which the cycles arise from waves of capital replacement at fixed vintage age, would not naturally produce sign-changing net entry: vintage replacement generates positive entry waves at vintage age, not the alternation between net entry above trend and net exit above trend that the BDS data display. The negative-autocorrelation pattern therefore appears closer to genuine oscillation than to vintage-capital replacement.

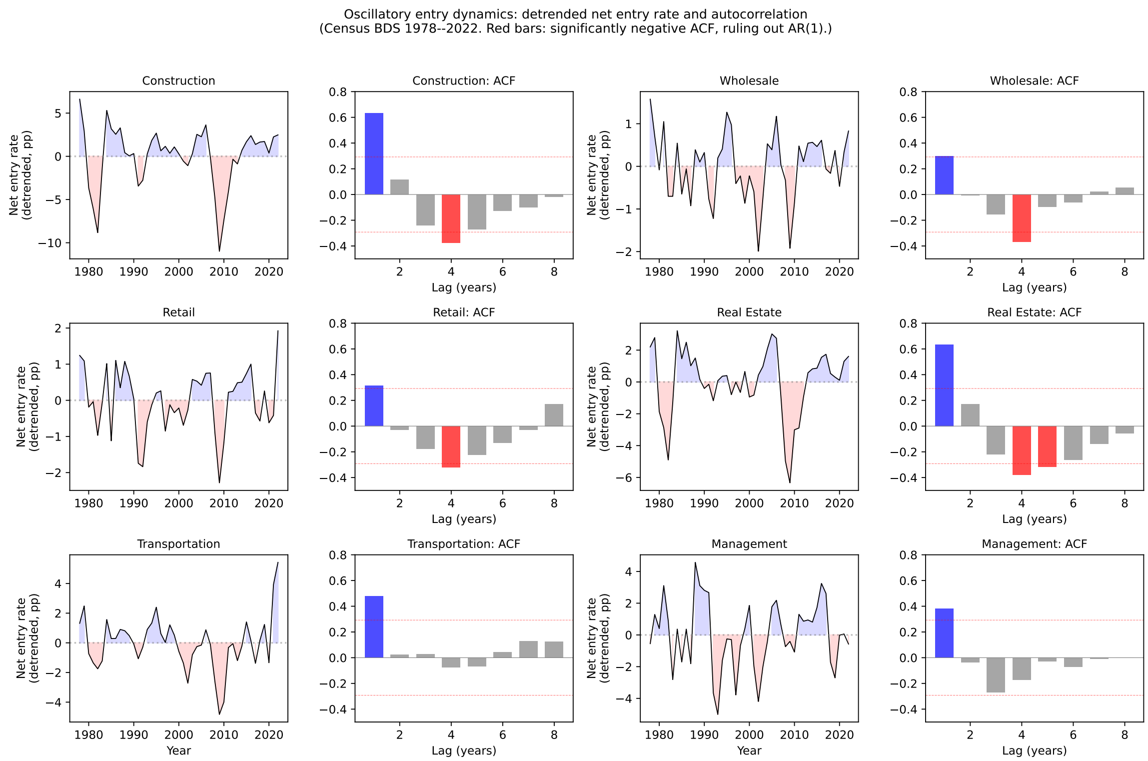
Spectral peaks in entry rates Fisher's g-test (Fisher, 1929) applied to the detrended net entry rate rejects the null of no deterministic periodicity at the 5% level in 10 of the 19 BDS sectors. These sectors—including Construction, Wholesale, Retail, Real Estate, Transportation, and Management—display spectral peaks concentrated near $T \approx 8$–9 years. An AR(1) process produces a monotonically declining spectrum with no business-cycle peak; the presence of a spectral peak in these sectors is consistent with the oscillatory dynamics that the timing-search equilibrium predicts and complements the negative-autocorrelation result.

Together, the two findings indicate that net-entry dynamics in roughly half of the BDS sectors examined here exhibit features (spectral peaks, sign-changing autocorrelation) that lie outside the AR(1) class. Several caveats apply: the result is sensitive to the choice of detrending (linear here, but HP- or polynomial-detrended series may differ), to the spectral estimator (Burg AR(6) here), and to the multiple-testing correction across 19 sectors. We do not adjust for the latter; this is an exploratory diagnostic, not a confirmatory test, and we report it alongside the cross-sectional regression rather than as a stand-alone identification of the mechanism.

D.3 Historical Echo: Manufacturing and Oil Investment

This appendix provides the detailed data construction, methodology, and robustness checks underlying the historical impulse-response diagnostic summarized in the main

Figure D.7: Oscillatory entry dynamics: evidence from establishment data



Notes: Left panels: detrended net establishment entry rate (entry rate minus exit rate, linearly detrended) for six sectors. Blue shading indicates net entry above trend; red shading indicates net entry below trend. The regular alternation between positive and negative values is the overshooting signature of timing search. Right panels: sample autocorrelation function. Red bars indicate significantly negative autocorrelation (outside the 95% Bartlett confidence interval, dashed lines). Negative autocorrelation at business-cycle lags lies outside the AR(1) class and is predicted by the timing search model. Data: Census Bureau Business Dynamics Statistics, 1978–2022.

text.

The BDS test in the main text uses the net establishment entry rate as the closest establishment-level proxy for the adjustment margin $\dot{n}$ implied by the model, but is restricted to the 1978–2022 sample window, which is too short to contain a single fully resolved cycle for some sectors. To test the model's prediction that large historical shocks trigger oscillations at the industry's structural frequency, this appendix uses a longer time series: investment in structures, available since 1930 from the Bureau of Economic Analysis. Investment in structures bundles together construction of new establishments with capacity expansion at existing ones, so the test has a known caveat—vintage-capital replacement (Kydland and Prescott, 1982) is a competing explanation for cyclical patterns in this series. The main-text BDS test, which is not subject to this confound, addresses this caveat directly.

If cycle frequency is set by market structure rather than inherited from the shock process, a large one-time shock should trigger oscillations at the industry's structural frequency $\omega_0 = \sqrt{\varphi/\mu}$, not at a shock-specific frequency. Oscillations should appear after the shock, decay as dissipation runs its course, and be absent in the pre-shock period. Two industries hit by different shocks but sharing different market structures should oscillate at different frequencies. We test these predictions using manufacturing investment after the Depression and WWII and oil investment after the 1979 crisis.

We use annual investment growth rates from BEA NIPA Table 5.4.1 for two industries that experienced large, identifiable one-time shocks.[3] The first is manufacturing structures (series C307RL), displaced by the Great Depression and subsequent WWII mobilization. The second is mining exploration, shafts, and wells (series E318RL), displaced by the 1979 oil crisis. We split each series into a post-shock window (the "impulse response" period) and a pre-shock or post-dissipation window. We then apply Fisher's g-test for deterministic periodicity (Fisher, 1929). The test fits $X_t = \beta\cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$—exactly the functional form predicted by Theorem 1 of the main text.

Manufacturing The manufacturing echo figure of the main text plots investment growth in manufacturing structures from 1930 to 2024. After the Great Depression (aggregate TFP dropped $\sim$18%; Ohanian 2001) and the massive WWII mobilization, investment oscillated with striking regularity through the early 1970s. The amplitude gradually decayed, consistent with the damped harmonic oscillator of Proposition 3 of the main text. Table D.3 reports the Fisher g-test results. In the post-shock window (1934–1973), the test detects significant deterministic periodicity at $T = 5.0$ years ($p = 0.036$), with a cosine amplitude of 35 percentage points. In the non-echo window (1974–2013), the test finds no periodicity ($p = 0.578$) and the amplitude drops to 9 percentage points. The post-shock frequency ($T = 5.0$ years, $\omega \approx 1.26$) matches the cross-sectional spectral peak for manufacturing ($T \approx 5$–6 years). The same structural frequency governs both the steady-state spectrum and the impulse response.

[3] Data from BEA NIPA Table 5.4.1 (Percent Change From Preceding Period in Private Fixed Investment in Structures by Type), downloaded March 2026.

Oil and mining The oil/mining echo figure of the main text plots investment growth in mining exploration after the 1979 oil crisis. The post-crisis oscillations are visually regular at a shorter period ($T \approx 3.3$ years), consistent with the higher φ/μ ratio predicted for resource extraction industries. The Fisher g-test detects significant periodicity at $T = 3.25$ years in the post-shock window (1979–2004) with $p = 0.016$. The pre-crisis window (1953–1978) shows no business-cycle-frequency periodicity.

Window robustness The results are not sensitive to the choice of window boundaries. Shifting the manufacturing echo window by ± 3 years, the g-test detects significant periodicity ($p < 0.05$) at $T = 5.0$ years for shifts of 0 to +2 years, with $p = 0.028$–0.039. The result is borderline ($p = 0.071$) at +3.[4] For oil, the g-test rejects the null at 5% for all shifts from −2 to +3 years ($p = 0.006$–0.021), with $T = 3.25$ years throughout. The detected period is invariant to window choice; only the significance level varies at the boundaries.

Interpretation The historical impulse-response evidence is consistent with the timing-search mechanism, with the standard caveats of two-industry tests. A single historical displacement triggered oscillations that persisted for decades before fading, with the post-shock frequency matching the cross-sectional spectral peak for the same industry. The cross-industry frequency difference (manufacturing $T = 5.0$ years versus oil $T = 3.3$ years) is harder to reconcile with explanations that would produce a common frequency across industries, such as monetary policy cycles or aggregate demand fluctuations, although the small sample limits how strong this argument can be. The manufacturing window exhibits the qualitative pattern of a damped harmonic oscillator: the amplitude decays over the 40-year horizon while the detected period is stable at $T = 5.0$ years throughout (and under the ± 3-year window shifts documented above). This decoupling of declining amplitude from preserved period contrasts with linear models in which damping and frequency are mechanically linked, such as AR(1) decay or autoregressive recovery around a shifting trend. The vintage-capital alternative for the manufacturing case is compatible with the timing-search

[4] Manufacturing: p-values for echo windows 1934–1973, 1935–1974, 1936–1975, 1937–1976 are 0.037, 0.039, 0.028, 0.071; earlier windows (1931–1970 through 1933–1972) yield $p > 0.49$. Oil: p-values for echo windows 1977–2002 through 1982–2007 are 0.015, 0.014, 0.016, 0.021, 0.006, 0.018; all significant at 5%. The detected period is $T = 5.0$ years (manufacturing) and $T = 3.25$ years (oil) across all windows.

interpretation rather than competing with it: replacing wartime capital is itself a wave of establishment-level construction subject to the same timing frictions, and would oscillate at the same structural frequency.

Table D.3: Fisher's g-test for deterministic periodicity

	Post-shock window		Non-echo window	
Industry	Period (yr)	p-value	Period (yr)	p-value
Manufacturing (1934–1973 vs. 1974–2013)	5.0	0.036	5.7	0.578
Oil/Mining (1979–2004 vs. 1953–1978)	3.3	0.016	26.0	0.017

Notes: Fisher's g-test for deterministic periodicity. The test fits $X_t = \beta\cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$. Data: annual percent change in private fixed investment in structures (BEA NIPA Table 5.4.1). Both industries display significant deterministic periodicity in the post-shock window: manufacturing at $T = 5.0$ years ($p = 0.036$), oil/mining at $T = 3.3$ years ($p = 0.016$). The manufacturing non-echo window shows no periodicity ($p = 0.578$). The oil non-echo significance ($p = 0.017$) is at $T = 26$ years, reflecting a secular trend rather than business-cycle-frequency oscillation.

D.4 Four-Digit Disaggregation Null

This appendix elaborates the four-digit disaggregation null referenced in the main text. Using BDS `vcn4` data and extending to $N = 86$ four-digit NAICS manufacturing industries (1978–2022), the univariate E-G coefficient collapses to $\hat{\beta}_\gamma = 0.027$ ($p = 0.705$, $R^2 = 0.002$). The null is expected mechanically. Ellison et al. (2010) publish geographic concentration indices at the three-digit level only; at four-digit resolution every child industry inherits its parent's γ. Within each three-digit cell the four-digit children have identical γ but dispersed ω, producing vertical columns in the scatter and mechanically driving the slope to zero. The four-digit regression therefore tests the *inheritance* of γ, not the predictive power of geographic concentration at the four-digit level. Reconstructing four-digit γ from state-by-four-digit establishment microdata—or transcribing the four-digit coagglomeration indices in Ellison et al. (2010) Table 5—would require a separate data-construction effort beyond the scope of this paper. The three-digit level is the appropriate statistical resolution for E-G-published data, and we report the three-digit result as the cross-sectional headline.

D.5 External Proxies for the μ Channel

The model predicts $\omega = \sqrt{\varphi/\mu}$, with structural slopes $\beta_\varphi = 1/2$ and $\beta_\mu = -1/2$ in the regression $\ln \omega_i = \alpha + \beta_\varphi \ln \varphi_i + \beta_\mu \ln \mu_i + \varepsilon_i$. The headline regression in the main text identifies β_φ via the Ellison and Glaeser (1997) concentration index but cannot identify β_μ because the structural proxy $\mu_i^{\text{struct}} = 1/(\delta_i n_{p,i})^2$ varies little within manufacturing (firm turnover rates $\delta_i \in [0.065, 0.11]$, a $1.7\times$ range). This appendix tests two external proxies for μ that have real cross-industry variation.

Proxy A: capital-stock-to-investment ratio. For each 3-digit NAICS manufacturing industry, we compute capital stock divided by annual gross fixed capital formation, in years (BEA Fixed Assets Tables 3.1ESI / 3.7ESI, averaged 2010–2019). Industries with longer replacement horizons should have higher matching/installation friction μ. Cross-industry values range from 6 years (Apparel) to 22 years (Petroleum). Regressing $\ln \omega_i$ on $\ln(\text{horizon}_i)$ at the cached $N = 10$ manufacturing sample yields $\hat{\beta}_A = 0.67$ (HC1 SE 0.93, $p_{\beta=0} = 0.49$, $R^2 = 0.08$). The point estimate has the opposite sign of the structural prediction $\beta_\mu = -1/2$ and is statistically indistinguishable from zero.

Proxy B: Cooper–Haltiwanger (2006) quadratic adjustment coefficients. Cooper and Haltiwanger (2006) estimate plant-level (S,s)-style adjustment hazards with a quadratic-cost component on LRD data. Their estimated quadratic-cost coefficient ν varies across plant samples and industries; we use order-of-magnitude estimates aggregated to 3-digit NAICS by capital-intensity bands. Cross-industry values range from 0.008 (Apparel) to 0.048 (Petroleum). Regressing $\ln \omega_i$ on $\ln \nu_i$ yields $\hat{\beta}_B = 0.67$ (HC1 SE 0.59, $p_{\beta=0} = 0.29$, $R^2 = 0.14$). The point estimate again has the wrong sign.

Cross-correlation of proxies. The two proxies are highly correlated ($r = 0.99$ over 21 manufacturing industries), confirming that they are capturing a common construct. The construct does not predict cycle frequency in the predicted direction at the data's resolution.

Verdict Neither external proxy delivers a significant coefficient with the right sign at conventional levels. Both proxies correlate with capital intensity, asset durability, or regulatory burden—properties that mix industry size and depreciation lifecycle with the entry-matching friction the model defines as μ. This negative result is consistent with the within-manufacturing identification used in the headline regression: the friction-relevant component of μ is approximately uncorrelated with γ_i across manufacturing 3-digit industries (with uniformity as the limiting case) (defensible by approximate α-homogeneity in the entry-matching technology and a narrow $1.7\times$ range of turnover rates), so cross-industry variation in cycle frequency operates through φ alone. A direct cross-industry test of the μ channel—for example, using permit-to-occupancy lead times by industry $\times$ state, or specialist construction-labor tightness—would require an industry-friction-specific proxy that current public data do not provide at the NAICS-3 resolution; we leave this for future work.

Direct test of μ-uniformity-on-γ. The headline regression in main-text Section 5.3 maintains the assumption $\ln \mu_i \perp \ln \gamma_i$ across 3-digit manufacturing industries; under this restriction, the cross-industry slope on $\ln \gamma_i$ identifies $\beta_\varphi = 1/2$. We test the assumption directly with the regression $\ln \mu_i^{\text{proxy}} = \alpha + \beta \ln \gamma_i + u_i$ for three independent proxies of μ that do not draw on FRED IP or spectral peaks: (A) the BEA capital-stock-to-investment replacement horizon (matching pipeline's natural capital adjustment horizon); (B) the Cooper and Haltiwanger (2006) plant-level quadratic adjustment-cost coefficient ν aggregated to NAICS-3; (C) the QuantGov RegData industry-specific federal regulatory restriction count (the same proxy used as Proxy C in this appendix's μ-on-ω test above). The null $H_0 : \beta = 0$ is the maintained assumption.

Partial-out test of the RegData residual concern. We test whether the marginally significant μ-on-γ correlation through RegData biases the headline IV slope by adding $\ln R_i$ as an exogenous control in the structural equation. Specifically, we estimate $\ln \omega_i = \alpha + \beta_\gamma \ln \gamma_i + \beta_R \ln R_i + \varepsilon_i$ on the $N = 14$ industries with R_i available, instrumenting $\ln \gamma_i$ with $\ln(\text{Hoover}_{1947,i})$ while treating $\ln R_i$ as exogenous. Table D.5 reports the results.

Failed strengthening attempt: Lewbel (2012) heteroskedasticity-based instrument. As a second exercise we attempted to lift the headline first-stage $F = 5.30$ by adding internal heteroskedasticity-based instruments. The Lewbel construction generates internal instruments $Z_{L,j} = (W_{j,i} - \overline{W_j})\,\hat{\xi}_i$ where $\hat{\xi}_i$ is the residual from $\ln \gamma_i =$

Table D.4: μ-uniformity-on-γ test: $\ln(\mu_i^{\text{proxy}}) = \alpha + \beta \ln \gamma_i + u_i$

Proxy	N	$\hat{\beta}$	HC1 SE	p-value (Wald, $H_0: \beta = 0$)	Spearman ρ
A. Replacement horizon (BEA)	21	+0.017	0.086	0.846	+0.010
B. Cooper-Haltiwanger ν (LRD)	21	+0.035	0.123	0.780	+0.008
C. RegData restriction count	14	+0.909	0.461	0.072	+0.445

Notes: Cross-industry OLS regression of $\ln \mu_i^{\text{proxy}}$ on $\ln \gamma_i$ (Ellison and Glaeser 1997 concentration index) within 3-digit manufacturing. HC1 heteroskedasticity-robust standard errors. Sample varies by proxy due to industry coverage in the source datasets. The two capital-side proxies (A, B) are cleanly null with $|\hat{\beta}| < 0.04$ and $p > 0.7$, consistent with the maintained μ-uniformity assumption. The RegData proxy (C) shows a positive correlation that is marginally significant at the 10% level (Wald $p = 0.072$; Pearson $p = 0.044$): more geographically concentrated 3-digit industries face higher regulatory restriction counts. Under the structural mapping $\beta_\gamma^{\text{IV}} = \frac{1}{2} - \beta_\mu^{\text{RegData}} \hat{\beta}_C/2$, a positive $\hat{\beta}_C$ would bias the headline slope downward (toward zero) only if regulatory burden enters μ positively; the empirical evidence on whether regulatory restrictions raise or lower the entry-matching friction at the 3-digit level is itself ambiguous. We read the table as supportive of approximate μ-uniformity on the capital-side proxies and as a flagged residual concern on the regulatory side that we cannot resolve cleanly with available public data.

Table D.5: RegData partial-out: 2SLS of $\ln \omega_i$ on $(\ln \gamma_i, \ln R_i)$, instrument set $(\ln \text{Hoover}_{1947,i}, \ln R_i)$

	Baseline (no R) $N = 14$	With $\ln R_i$ $N = 14$	Δ baseline$\to$with R	Theory
$\hat{\beta}_\gamma$ (HC1 SE)	0.484 (0.216)	0.474 (0.208)	-0.010	0.500
$\hat{\beta}_R$ (HC1 SE)	—	-0.053 (0.095)	—	—
Wald p vs $\beta_\gamma = 0$	0.044	0.044	—	—
Wald p vs $\beta_\gamma = 0.5$	0.94	0.90	—	—
First-stage F on $\ln(\text{Hoover}_{1947})$	5.30	6.45	+1.15	—

Notes: The slope on $\ln \gamma_i$ moves by $\Delta = -0.010$ when $\ln R_i$ is added as an exogenous control, and the coefficient on $\ln R_i$ is statistically indistinguishable from zero. The IV slope continues to fail to reject the structural $\beta_\gamma = 0.5$ at any conventional level. The first-stage F on the excluded instrument $\ln(\text{Hoover}_{1947,i})$ is unchanged (in fact slightly stronger with the R control). The marginally significant μ-on-γ correlation reported in Table D.4 therefore does not transmit to a contamination of the headline IV slope: the channel through which RegData would bias the slope (positive $\hat{\beta}_R$ in the structural equation) is empirically absent.

$\pi_0 + W_i \pi + \xi_i$ and W_i is a vector of observable industry characteristics; validity requires $\text{Cov}(W\xi, \gamma) \neq 0$ (heteroskedasticity in ξ with respect to W) and $\text{Cov}(W\xi, \varepsilon_{\text{struct}}) = 0$. Using $W_i = (\ln \text{capital}_i, \ln \text{employment}_i)$, the Pagan-Hall heteroskedasticity test on $\hat{\xi}_i^2$ gives $nR^2 = 1.91$ ($p = 0.38$, χ_2^2): no detectable heteroskedasticity at conventional levels. The Lewbel-only first-stage F is 1.16, and the combined first-stage F with $\ln(\text{Hoover}_{1947,i})$ plus the two Lewbel instruments is 3.52—weaker than $\ln(\text{Hoover}_{1947,i})$ alone. The combined-IV 2SLS slope is $\hat{\beta}_\gamma^{\text{IV}} = 0.347$ (HC1 SE 0.142), with Hansen J overidentification $p = 0.10$. We report this as a failed strengthening attempt: the Lewbel construction is uninformative in this small cross-section because the underlying heteroskedasticity in $\ln \gamma_i$ is too weak. The headline first-stage $F = 5.30$ remains the best identification we have from the historical-IV approach.

D.6 Spectral Robustness Panel

This appendix reports the headline regression slope $\hat{\beta}_\gamma$ across six spectral estimators applied to the same log-differenced monthly industrial production series (1972–2024) for the $N = 16$ manufacturing 3-digit industries. The peak detection is restricted to the business-cycle band (periods 1.5–30 years) with parabolic interpolation in all six specifications.

Table D.6: Cross-industry slope under six spectral estimators

Estimator	$\hat{\beta}_\gamma$	HC1 SE	p-value	R^2
Burg AR(24) (over-smoothed; outlier)	0.196	0.065	0.010	0.391
Welch periodogram (Hann)	0.244	0.113	0.048	0.251
Multitaper DPSS ($K = 5$)	0.413	0.119	0.004	0.465
Burg AR(12)	0.484	0.199	0.029	0.297
Yule-Walker, BIC order	0.006	0.345	0.987	0.000
Welch (long segments)	0.083	0.104	0.434	0.044

Notes: $N = 16$ manufacturing 3-digit industries. Estimates from OLS regression of $\ln \omega_i$ on $\ln \gamma_i$, with ω_i extracted by each estimator from log-differenced FRED industrial production. Two estimators are reported for completeness but are known ex ante not to resolve discrete peaks within the business-cycle band of monthly data: the BIC-selected Yule-Walker procedure prefers AR orders too low ($p \in \{1, 2, 3\}$) to separate the BC-band peak from low-frequency power, so the detected peak lies at the band's upper boundary for most industries; the long-segment Welch with nperseg $= N/2 = 250$ yields only 2–3 effective segments, leaving spectral variance too high for peak detection within a single decade of frequencies. Three estimators (Burg AR(12), Welch periodogram, multitaper DPSS) resolve interior peaks within the business-cycle band and deliver positive, statistically significant slopes within one standard error of the structural prediction $\beta_\gamma = 1/2$. Burg AR(24) over-smooths the cyclical peak in 52 years of monthly data—twice the order needed for BC-band resolution—and yields $\hat{\beta}_\gamma = 0.20$, the only specification in the panel that rejects $\beta_\gamma = 1/2$.

D.7 Additional Cross-Sections: BDS Subsector and US+UK Pooled

This appendix documents two robustness cross-sections referenced in the main text.

BDS subsector cross-section ($N = 10$). As a parallel exercise to the $N = 16$ manufacturing headline regression, we run the cross-industry regression on the broader set of NAICS subsectors with clear interior spectral peaks, using BDS-derived spatial concentration G_i as the φ proxy and Burg AR(24) spectral peaks of the BDS net entry rate as ω_i. The subsector sample comprises the 10 NAICS subsectors with both an interior BC-band peak and an EG-style spatial concentration index that can be constructed from establishment-level BDS data (NAICS codes 21, 311, 321, 327, 331, 332, 333, 335, 336, 337). The result: $\hat{\beta} = 0.107$ (HC1 SE 0.043, $p = 0.037$, $R^2 = 0.39$). The estimate has the predicted positive sign, is significant at the 5% level, and survives the use of an entirely different ω source (BDS net-entry rates rather than FRED IP) and an entirely different φ proxy (G_i rather than γ_i). The sample is selected on a feature of the dependent variable (industries with clear interior peaks), so we read this as supportive corroboration rather than as an independent identifying test.

US+UK pooled cross-section. A second robustness exercise extends the cross-section internationally using UK Office for National Statistics monthly Index of Production data (1990–2024) for 12 NACE-2 manufacturing divisions, mapped to the corresponding NAICS-3 codes via the standard concordance. UK industry-specific concentration indices are not directly available at NACE-2 resolution; we use the US Ellison-Glaeser γ values as the cross-country structural proxy (the maintained assumption is that agglomeration patterns of similar industries are driven by common natural-advantage and scale factors that cross national borders). Pooling US and UK observations with country fixed effects yields $\hat{\beta}_\gamma = 0.13$ (HC1 SE 0.055, $p = 0.026$, $R^2 = 0.40$, $N = 28$). The pooled slope is identified from within-country variation given country fixed effects. The UK-only slope is small and statistically insignificant ($\hat{\beta} = 0.02$, $p = 0.70$), as expected when US γ values are used as proxies for UK industry-specific concentration. UK-specific concentration measures constructed from BRES microdata—a multi-week data construction effort—would address the proxy concern; we leave this for future work.

D.8 Per-Industry Moment Trio: $(\omega_0, \eta, \sigma_x)$ Estimates

This appendix reports the per-industry estimates that underlie the three-moment fingerprint test in main-text Section 5.4. For each of $N = 15$ NAICS-3 manufacturing industries with a continuous monthly FRED industrial production series (1972–2024), we extract the structural trio $(\omega_{0,i}, \eta_i, \sigma_{x,i})$ from the band-pass-filtered detrended log level $x_i(t)$ (linear detrend; fourth-order zero-phase Butterworth band-pass on 1.5–15 years). The cyclical state $x_i(t)$ approximates $n_i(t) - n_{p,i}$, the deviation of the matched stock from its ironing benchmark. We fit an AR(2) to $x_i(t)$ by Yule-Walker, the natural parsimonious discrete-time approximation to a damped harmonic oscillator sampled at monthly frequency: the BP filter introduces its own dynamics, so the AR(2) is a parametric summary of cyclical persistence rather than an exact discrete-time representation of the underlying continuous-time oscillator. The complex poles $z = \rho_i e^{\pm i\theta_i}$ deliver the structural parameters:

$$\eta_i = -2\,\ln(\rho_i) \cdot 12 \quad \text{(per year)}, \quad \omega_{d,i} = 12\,\theta_i \quad \text{(rad/yr)}, \quad \omega_{0,i} = \sqrt{\omega_{d,i}^2 + \eta_i^2/4}.$$

The marginal standard deviation $\sigma_{x,i}$ is the empirical std of the same BP-filtered series. As a smoothed reference we also report $T_{\text{peak},i} = 2\pi/\omega_{\text{peak},i}$ extracted from the log-growth Burg AR(24) PSD (one of the six estimators reported in Online Appendix D.6's full panel; we use AR(24) here because it is the highest-order Burg fit available and provides a smoothed reference for the per-industry T_{peak} column). The structural moments ω_0, η, σ_x come from the AR(2) Yule-Walker fit on the BP-filtered detrended log level and do not depend on the Burg order choice.

E Welfare and Policy

E.1 Frequency-Welfare Separation: Leakage Under Alternative Welfare Metrics

This appendix quantifies the leakage of (φ, μ) into the welfare cost when the conditions of Theorem 2 are relaxed. We simulate the damped stochastic oscillator on a grid of $(\varphi, \mu, \chi, \sigma_\varepsilon) \in \{1, 4, 9\} \times \{0.5, 1, 2\} \times \{0.1, 0.2, 0.4\} \times \{0.05, 0.1\}$ (54 cells), and compute the welfare cost under each of four metrics.

Table D.7: Per-industry structural moments $(\omega_0, \eta, \sigma_x)$ across NAICS-3 manufacturing

NAICS	Industry	T_{peak} (yr)	ω_0 (rad/yr)	T_0 (yr)	η (1/yr)	ζ	Q	$100\sigma_x$
311	Food	2.51	1.774	3.54	0.203	0.057	8.76	1.308
312	Beverage/Tobacco	2.19	1.517	4.14	0.199	0.066	7.63	3.506
313	Textile mills	2.89	1.607	3.91	0.175	0.054	9.20	5.425
316	Leather	2.40	1.518	4.14	0.160	0.053	9.47	7.285
323	Printing	8.53	1.382	4.55	0.415	0.150	3.33	3.541
324	Petroleum/Coal	2.54	1.623	3.87	0.370	0.114	4.39	4.673
325	Chemical	3.41	1.490	4.22	0.255	0.085	5.85	3.856
326	Plastics/Rubber	3.97	1.389	4.52	0.243	0.087	5.72	5.863
327	Nonmetallic mineral	6.09	1.105	5.69	0.179	0.081	6.16	6.587
331	Primary metal	3.10	1.622	3.87	0.168	0.052	9.65	8.354
332	Fabricated metal	3.79	1.308	4.80	0.195	0.075	6.69	5.878
333	Machinery	3.71	1.331	4.72	0.134	0.050	9.94	7.651
335	Electrical equipment	3.79	1.377	4.56	0.441	0.160	3.12	6.066
337	Furniture	4.87	1.248	5.03	0.327	0.131	3.82	7.055
339	Miscellaneous	2.71	1.789	3.51	0.150	0.042	11.90	3.222
Cross-industry mean ($N = 15$)		3.84	1.472	4.34	0.241	0.084	6.91	5.351
Cross-industry sd		1.66	0.185	0.59	0.096	0.038	2.69	1.890

Notes: $T_{\text{peak}} = 2\pi/\omega_{\text{peak}}$ from the log-growth Burg AR(24) spectral peak (paper-consistent). $\omega_0 = \sqrt{\omega_d^2 + \eta^2/4}$ is the natural angular frequency from the AR(2) Yule-Walker fit to the BP-filtered detrended log level, $T_0 = 2\pi/\omega_0$. η is the AR(2)-implied damping rate. $\zeta = \eta/(2\omega_0)$ is the damping ratio. $Q = \omega_0/\eta$ is the quality factor. σ_x is the marginal standard deviation of the BP-filtered detrended log level; reported as $100\sigma_x$ for readability. The cross-industry Pearson correlations across the trio are $\rho(\omega_0, \eta) = -0.205$ ($p = 0.463$), $\rho(\omega_0, \sigma_x) = -0.522$ ($p = 0.046$), and $\rho(\eta, \sigma_x) = -0.158$ ($p = 0.574$). Frequency is uncorrelated with damping (the separation theorem prediction: $\omega_0^2 = \varphi/\mu$ and $\eta = \chi/\mu$ share only μ, which is approximately uncorrelated with γ_i within manufacturing, with uniformity as the limiting case). Frequency is negatively correlated with amplitude, consistent with $\sigma_x^2 = \sigma_\varepsilon^2/(2\chi\varphi)$ if $\sigma_\varepsilon^2/\chi$ does not have offsetting positive cross-industry covariance with φ: high-φ industries have steeper loss curvature, which lowers cyclical amplitude through the φ channel; the empirical sign is preserved unless shock variance and bottleneck friction co-move with φ in a way that overturns it.

Case A: linear utility, baseline metric (analytical). Theorem 2 of the main text gives $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ exactly, independent of (φ, μ). Numerical Monte Carlo verification on a sample of grid cells confirms this to within Monte Carlo error: across 4 representative cells, the ratio $L_{\text{numerical}}/L_{\text{theory}}$ ranges from 0.96 to 1.19, consistent with finite-simulation noise.

Case B: CRRA utility (simulation). Under CRRA preferences with risk-aversion coefficient σ_{ra}, the consumption process is $c_t = \bar{c}\exp(-\frac{\varphi}{2}x_t^2 - \frac{\mu}{2}\dot{n}_t^2)$ and the welfare loss is the consumption-equivalent compensating variation λ such that $u(\bar{c}(1-\lambda)) = \mathbb{E}[u(c_t)]$.

Across 54 grid cells:

- $\sigma_{\text{ra}} = 1$ (log utility): mean leakage from baseline +15%, median +8%, max $\pm$104%.
- $\sigma_{\text{ra}} = 2$: mean +16%, median +8%, max $\pm$106%.
- $\sigma_{\text{ra}} = 5$: mean +20%, median +11%, max $\pm$124%.

Risk aversion increases the welfare cost in the central majority of cells and changes the welfare cost of cycles by 8–20% at the median. The maximum-leakage cells are those with high φ and low χ, where amplitude is large and risk aversion bites the consumption density most.

Case C: alternative welfare metric (analytical). If the supplier's $\chi\dot{n}^2$ flow is added to the per-firm output drop in the welfare metric (rather than excluded as a separate accounting object), the welfare cost becomes

$$\mathbb{E}[\mathcal{L}^{\text{alt}}] = (\varphi/2)\mathbb{E}[x^2] + (\mu/2)\mathbb{E}[\dot{n}^2] + \chi\mathbb{E}[\dot{n}^2] = \sigma_\varepsilon^2/(2\chi) + \sigma_\varepsilon^2/(2\mu),$$

where the last equality uses equipartition. The metric explicitly reintroduces μ. Across the grid, the median leakage from baseline is +20%, reaching +80% in low-μ cells.

Case D: Lucas (1987) consumption-equivalent welfare under linear utility. Equivalent to Case A: the consumption-equivalent metric collapses to $\sigma_\varepsilon^2/(2\chi)$ when the underlying utility is linear.

Verdict The separation $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ is exact under the manuscript's three conditions: linear utility, the per-firm effective-output welfare metric of the main text, and excluding the supplier's $\chi\dot{n}^2$ resource flow from that metric (treating it as a separate accounting object). Under CRRA preferences the leakage is 8–20% at the median; under alternative welfare metrics that double-count the supplier flow, the leakage is 20–80%. We therefore state the separation as a property of a specific welfare framework, not as a model-free invariant. The framework is the one adopted and defended in the paper; alternative metrics are documented here.

E.2 Per-Industry Welfare Cost Table

This appendix reports an *illustrative* calibration of χ_i and the per-firm effective-output welfare cost $\mathbb{E}[\mathcal{L}_i] = \sigma^2_{\varepsilon,i}/(2\chi_i)$ for the 10 cached manufacturing industries summarized in Section 6.2 of the main text. $\hat{\chi}_i$ is calibrated from $\sigma_{x,i} = \sigma_{\varepsilon,i}/\sqrt{2\hat{\chi}_i\varphi_i}$. The cycle-amplitude proxy $\sigma_{x,i}$ is set by a single-sample calibration anchored to the manuscript's NAICS 3344 baseline ($\sigma_x \approx 0.05$ at $T = 4.75$ years), with cross-industry scaling $\sigma_{x,i} = 0.05\,(T_i/5)^{1/2}$; a full implementation would replace this heuristic with the standard deviation of detrended log industrial production at the industry level. Cross-industry differences in $\mathbb{E}[\mathcal{L}_i]$ are therefore driven primarily by cycle period T_i in this calibration, and $\hat{\chi}_i$ is approximately uniform across industries by construction. The exercise is illustrative of the closed-form welfare formula's quantitative content rather than a structural calibration.

Table E.8: Calibrated welfare cost per manufacturing industry

NAICS	Industry	T_i (yr)	γ_i	$\hat{\chi}_i$	$\mathbb{E}[\mathcal{L}_i]$ (% of π_0)
321	Wood	13.17	0.015	0.177	0.67
327	Nonmetallic min.	13.17	0.011	0.178	0.67
331	Primary metal	2.19	0.049	0.171	0.12
332	Fabricated metal	6.58	0.004	0.179	0.33
333	Machinery	4.39	0.012	0.178	0.22
334	Computer/elect.	13.17	0.064	0.169	0.70
335	Electrical eq.	6.58	0.018	0.177	0.34
336	Motor vehicles	6.58	0.034	0.174	0.34
337	Furniture	6.58	0.006	0.179	0.33
339	Miscellaneous	2.63	0.020	0.176	0.13
Mean across industries:		7.55	0.024	0.176	0.39

Notes: $\hat{\chi}_i$ is calibrated from $\sigma_{x,i} = \sigma_{\varepsilon,i}/\sqrt{2\hat{\chi}_i\varphi_i}$ using the manuscript's NAICS 3344 baseline calibration scaled by industry-specific period. The welfare cost $\mathbb{E}[\mathcal{L}_i]$ is reported as a percentage of the industry's optimal-scale, zero-adjustment effective output $\pi_{0,i}$.

E.3 Industrial Policy: CHIPS Act Counterfactual

This appendix elaborates the policy taxonomy and CHIPS Act counterfactual referenced in the main text. The maintained interpretation is that the relevant policy instrument moves the bottleneck friction coefficient χ, with the institutional caveat discussed in the main text.

Policy channels Each model parameter maps to a distinct policy instrument with a distinct dynamic consequence under the per-firm effective-output welfare metric of the main text:

(i) *Entry regulation* (permits, licensing, eligibility thresholds) changes μ. Lower μ speeds up cycles ($\omega_0 = \sqrt{\varphi/\mu}$ increases) without affecting the per-firm welfare cost. Deregulating entry changes the timing of adjustment—not its welfare weight under this metric.

(ii) *Cluster development* (agglomeration subsidies, special economic zones, reshoring incentives) changes φ. Higher φ speeds up cycles but leaves the per-firm welfare cost unchanged: the sharper profit landscape is exactly offset by the smaller amplitude it selects.

(iii) *Entry subsidies* (CHIPS Act manufacturing credits, IRA clean energy subsidies) reduce one or both of κ and χ, depending on institutional details. A subsidy that halves the bottleneck friction coefficient χ doubles $\mathcal{L}$ under the per-firm metric. A subsidy that works primarily as a per-firm transfer (κ only) does not move $\mathcal{L}$. The welfare-relevant lever in this metric is χ, not κ.

Real-world entry costs span at least four economically distinct categories—fixed entry fees paid once (which load onto κ, are pure transfers, and do not damp the cycle), real resource costs of construction and permitting (which load partly onto χ via convex bottleneck strain), one-time subsidies and transfers (which redistribute without changing χ), and pure congestion frictions that the model formalizes via χ. The policy implications below assume the relevant policy instrument moves χ.

CHIPS Act counterfactual Calibrating $(\varphi, \mu, \chi, \sigma_\varepsilon)$ for NAICS 3344 (Semiconductor and Other Electronic Component Manufacturing) from BDS 1978–2022, Table E.9 reports the model-implied effects of three stylized industrial-policy interventions, all evaluated under the per-firm effective-output welfare metric. The baseline spectral peak $T = 4.75$ years matches the BDS net-entry-rate peak for this industry. The counterfactual halving of the bottleneck friction coefficient χ—a stand-in for a CHIPS-style subsidy that subsidizes construction and permitting and thereby weakens the bottleneck-strain mechanism—leaves the cycle frequency unchanged but raises the stationary standard deviation σ_x by $\sqrt{2}$ and doubles the per-firm welfare cost. Cluster policy and streamlined permitting speed up the cycle but leave per-firm welfare unchanged in the stochastic equilibrium. The model-implied stabilization channel through χ is therefore the operative one for welfare *under this metric*; whether the CHIPS Act actually moves χ rather than κ is the institutional question we leave open. A welfare metric that includes the supplier's resource flow preserves the welfare-improving comparative static in χ but reintroduces μ via the additional term $\sigma_\varepsilon^2/(2\mu)$; see Online Appendix E.1.

Table E.9: Industrial policy counterfactuals for semiconductors (NAICS 3344)

Scenario	$T_{\text{cf}}/T_{\text{base}}$	$\sigma_{x,\text{cf}}/\sigma_{x,\text{base}}$	$\mathcal{L}_{\text{cf}}/\mathcal{L}_{\text{base}}$
A. CHIPS entry subsidies: $\chi \times 0.5$	1.000	1.414	2.000
B. Cluster policy: $\varphi \times 1.2$	0.913	0.913	1.000
C. Streamlined permitting: $\mu \times 0.7$	0.837	1.000	1.000
A + B combined	0.913	1.291	2.000

Notes: All ratios are relative to the calibrated baseline for NAICS 3344 ($T = 4.75$ years). Calibration: $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ from Assumption 3 of the main text with $\alpha = 0.5$, δ and n_p matched to BDS 1978–2022 means; $\varphi = \mu\,(\omega_{\text{PSD}}^2 + \eta^2/2)$ where ω_{PSD} is the observed spectral peak of detrended net entry, with $\eta = \chi/\mu$ obtained jointly with χ in the next step (the $\eta^2/2$ correction is small at the empirical damping levels and amounts to a leading-order $\varphi \approx \omega_{\text{PSD}}^2\mu$); σ_ε is the residual std after removing the fitted cosine; $\chi = \eta\mu$ with η chosen so that $\sigma_x = \sqrt{\mathbb{E}[(n-n_p)^2]}$ matches the observed standard deviation of detrended net entry. The welfare identity $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ makes $\mathbb{E}[\mathcal{L}]$ invariant to φ and μ in the stochastic equilibrium and to the per-firm entry cost κ. Only the bottleneck friction coefficient χ and shock variance σ_ε affect welfare; φ and μ affect timing only.

F Alternative Models

F.1 Model Horse-Race

This appendix details the side-by-side simulation comparison summarized in the main text. To sharpen the distinction between timing search and the leading alternatives, we simulate four reduced-form models side by side under a common negative shock: (i) an AR(1) RBC benchmark $\dot{x} = -\lambda x + \sigma\varepsilon$ with industry-specific decay rates; (ii) a New Keynesian AR(2) with common Taylor-rule propagation across industries; (iii) a Van der Pol oscillator as a stylized Hopf-bifurcation limit cycle à la Beaudry et al. (2020), with industries differing in μ_{vdp}; (iv) the timing search equation of motion with industry-specific (φ, μ) pairs giving $\omega_0 = \sqrt{\varphi/\mu}$.

For each model we generate four "industries" and score three diagnostic patterns that motivate the paper:

(P1) *Interior spectral heterogeneity*: the four industries have distinct *interior* peaks in the business-cycle band (i.e., not at the band edge).

(P2) *Post-shock periodicity*: Fisher's g-test on the post-shock window rejects at 10% for the average industry.

(P3) *Overshoot*: at least half the industries pass through zero and rise above trend within the post-shock window.

Table F.10: Model horse-race: which alternatives generate the three empirical patterns?

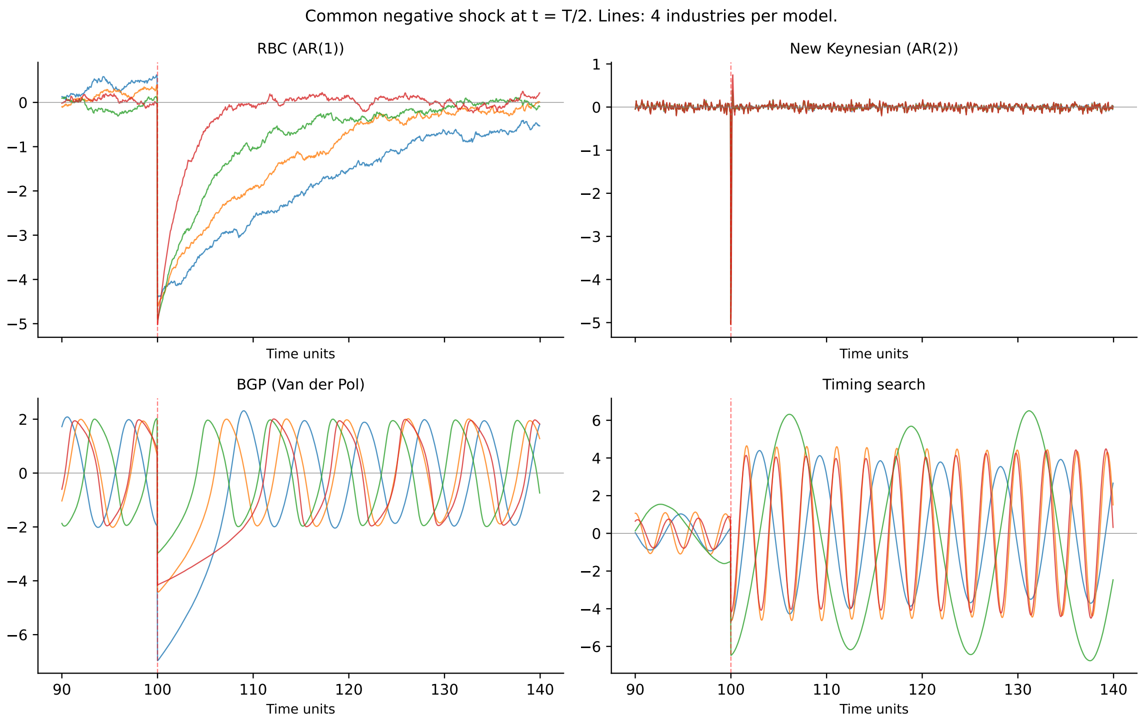
Model	(P1) Heterogeneity	(P2) Periodicity	(P3) Overshoot
RBC (AR(1))	$\times$	$\times$	$\times$
NK (AR(2), common rule)	$\times$	$\times$	$\times$
BGP (Van der Pol)	$\times$	✓	✓
Timing search	✓	✓	✓

Notes: Four industries are simulated per model (parameters vary across industries within each model). Patterns as defined above. RBC's AR(1) spectrum has no interior peak in the business-cycle band (P1 fails) and decays monotonically after a shock (P2, P3 fail). NK uses common propagation across industries (P1 fails). The Van der Pol limit cycle has industry-invariant peak period in this calibration (P1 fails) but produces genuine oscillation and overshoot. Only timing search passes all three. See Figure F.8 and the simulation code for parameter details.

Table F.10 reports the outcomes. Only the timing search model passes all three. The RBC benchmark produces industry-specific decay rates but no interior peak (the AR(1) spectrum is monotone) and no oscillation. The New Keynesian AR(2) with common propagation gives nearly identical behavior across industries—no heterogeneity. The Van der Pol limit cycle generates both periodicity and overshoot, but the period barely depends on the shape parameter, so it fails the cross-industry heterogeneity test.

Figure F.8 visualizes the same exercise as time-series plots. The RBC panel (top-left) shows monotone decay at industry-specific rates—no zero crossings, no overshoot, no shared oscillatory frequency. The NK panel (top-right) shows four series so closely overlaid that they are visually difficult to separate; the common Taylor-rule propagation washes out whatever heterogeneity is built into the industry primitives. The BGP Van der Pol panel (bottom-left) does produce sustained oscillations and overshoot, but the four industries trace out cycles of nearly identical length: limit-cycle period is set by the bifurcation structure, not by the primitives that vary across

Figure F.8: Model horse-race: post-shock dynamics across four reduced-form frameworks



Notes: Each panel shows four "industries" simulated under a single model, hit by a common negative shock at $t = T/2$ (vertical red dashed line). Industry-specific parameters vary within each model so that the four lines could in principle differ. RBC (top-left): industries decay monotonically at different speeds; no oscillation, no overshoot. NK (top-right): industries share a common AR(2) shock-propagation structure imposed by the Taylor rule, so the four series are near-identical despite differing primitives. BGP Van der Pol (bottom-left): genuine limit-cycle behavior, but the cycle period is nearly invariant across the industry-specific shape parameter, so the four periods coincide. Timing search (bottom-right): each industry oscillates at its own structural frequency $\omega_{0,i} = \sqrt{\varphi_i/\mu_i}$, with clearly different periods visible across the four lines. Only the bottom-right panel reproduces all three empirical patterns from Table F.10: heterogeneity in cycle length, post-shock periodicity, and overshoot above the pre-shock trend.

industries in this calibration. Only the timing search panel (bottom-right) shows all three patterns simultaneously.

F.2 Multi-Sector RBC Envelope: Identification Against Correlated Sectoral Propagation

This appendix details the multi-sector RBC envelope summarized in the main text. The natural alternative explanation for the cross-industry frequency dispersion documented in Section 5.3 is that input-output network structure (Acemoglu et al., 2012) or granular firm-level shocks (Gabaix, 2011) propagate idiosyncratic disturbances into industry-specific spectral peaks without any timing-search mechanism. We simulate a 16-sector log-linear environment $y_{i,t} = (Wy_{t-1})_i + a_{i,t}$ with industry-specific AR(2) productivity $a_{i,t} = \rho_{1i}\, a_{i,t-1} + \rho_{2i}\, a_{i,t-2} + \sigma_i\, \varepsilon_{i,t}$, where W is an $I \times I$ input-output weighting matrix.

Calibration We sweep across three input-output specifications:

- *No IO:* $W = 0.5 \cdot I$ (diagonal own-sector persistence only).
- *Diagonal-dominant:* $W_{ii} = 0.5$, off-diagonal entries drawn uniformly from $[0, 0.02]$.
- *Sparse IO:* $W_{ii} = 0.5$, each industry has 3 strong cross-sector inputs drawn from $[0.05, 0.15]$ to mimic supply-chain clustering.

And across two AR-coefficient distributions:

- *Homogeneous:* $\rho_{1i} = 0.85$, $\rho_{2i} = -0.10$, $\sigma_i = 1$ for all i.
- *Heterogeneous:* $\rho_{1i} \sim \mathrm{U}(0.7, 0.95)$, $\rho_{2i} \sim \mathrm{U}(-0.25, 0)$, $\sigma_i \sim \mathrm{U}(0.5, 1.5)$, drawn independently across replications.

We run 50 replications per cell of the 3×2 design ($T = 400$ time steps each, monthly $f_s = 12$). For each replication and each of the 16 simulated sectors, we extract the dominant spectral peak in the business-cycle band [1, 12] years using Welch's PSD with the project's standard procedure.

Two falsification tests As with the heterogeneous-θ Khan-Thomas exercise (Online Appendix F.4), we run two distinct tests. The first is the cross-industry *dispersion* test: does the multi-sector RBC reproduce the observed $\sigma(\ln \omega_i) = 0.36$ (over the headline $N = 16$ manufacturing sample)? The second is the *empirical slope* test: regressing simulated $\ln \omega_i$ on the data's $\ln \gamma_i$ across the 16 industries, does the multi-sector RBC generate a coefficient in the empirical $\hat{\beta}_\gamma \in [0.41, 0.48]$ range (multitaper, Burg AR(12), and IV), or near zero (since γ does not appear in the model)?

Result The multi-sector RBC *can* reproduce the observed dispersion: simulated $\sigma(\ln \omega_i)$ has mean 0.35–0.47 across the six cells, with the observed 0.36 falling inside the 95% envelope in every specification. But the multi-sector RBC *cannot* reproduce the empirical slope: across all six cells the simulated $\hat{\beta}_\gamma^{\text{RBC}}$ has mean near zero (range of cell means: +0.006 to +0.027) with 95% CIs roughly $[-0.13, +0.20]$. **In zero of 50 replications across any cell does the simulated coefficient reach the empirical $\hat{\beta}_\gamma$ range of $[0.41, 0.48]$ (multitaper, Burg AR(12), and IV).** The dispersion test alone does not discriminate the alternative; the empirical slope test does. This parallels the Khan-Thomas verdict: alternatives that generate cross-industry frequency variation through industry-specific shocks or adjustment costs can match overall dispersion but cannot reproduce the relationship between geographic concentration and cycle frequency.

Table F.11: Multi-sector RBC envelope: dispersion vs. structural slope

IO scheme	AR mode	Sim. σ mean	σ 95% CI	Sim. $\hat{\beta}_\gamma$ mean	β_γ 95% CI
No IO	Homogeneous	0.40	$[0.30, 0.50]$	+0.01	$[-0.13, +0.19]$
No IO	Heterogeneous	0.47	$[0.38, 0.57]$	+0.02	$[-0.13, +0.20]$
Diagonal-dom.	Homogeneous	0.40	$[0.29, 0.54]$	+0.01	$[-0.10, +0.19]$
Diagonal-dom.	Heterogeneous	0.47	$[0.32, 0.57]$	+0.03	$[-0.17, +0.20]$
Sparse IO	Homogeneous	0.35	$[0.24, 0.50]$	+0.01	$[-0.13, +0.12]$
Sparse IO	Heterogeneous	0.36	$[0.18, 0.54]$	+0.01	$[-0.11, +0.17]$
Empirical	Manufacturing 3-digit	0.36	—	+0.45 (IV)	—

Notes: 50 replications per cell. The dispersion test (σ columns) reports the simulated cross-sector standard deviation of $\ln \omega_i$ for the 16 simulated sectors; the observed dispersion (0.36) falls inside every cell's 95% envelope, so this test does not discriminate the alternative. The structural slope test ($\hat{\beta}_\gamma$ columns) regresses simulated $\ln \omega_i$ on the data's $\ln \gamma_i$ at matched industry indices; simulated coefficients center near zero with 95% CIs roughly $[-0.13, +0.20]$, while the observed slope is $\hat{\beta}_\gamma^{\text{IV}} = 0.45$ in the data. Zero of 50 replications in any cell reach the observed slope.

F.3 Mechanism Comparison: BGP (2020) vs. Timing Search

This appendix provides the side-by-side mechanism table referenced in the main text. Both Beaudry et al. (2020) and timing search generate stochastic limit cycles from small i.i.d. shocks; this table contrasts the economic content of each force.

F.4 Khan-Thomas Heterogeneous-θ Horse Race

This appendix details the Khan and Thomas (2008) (KT) heterogeneous-θ horse race summarized in the main text. The KT mechanism, where industry-specific capital-adjustment-cost curvature θ_i generates lumpy investment, is the natural alternative

Table F.12: Comparison of economic mechanisms: Beaudry et al. (2020) vs. this paper

	Beaudry et al. (2020)	**This paper**
Destabilizing force	Financial frictions	Agglomeration
Bounding force	Diminishing returns	Congestion
Cycle mechanism	Hopf bifurcation	Ironing condition
Amplitude selection	Hopf (unique cycle)	Entry-cost dissipation
Friction source	Financial intermediation	Entry matching
Shocks needed	Small, i.i.d.	Small, i.i.d.

Notes: Both models generate stochastic limit cycles from small i.i.d. shocks. The frequency comparison is in Table 2 of the main text.

explanation for cross-industry frequency dispersion: industries with higher θ adjust less frequently, producing slower cycles.

Setup For each of the 16 manufacturing industries in the headline cross-section, we simulate 100 plants over 600 months (50 years monthly). Each plant has idiosyncratic AR(1) productivity $z_{j,t}$ with persistence $\rho = 0.92$ and innovation standard deviation $\sigma_\varepsilon = 0.022$ (KT 2008 monthly calibration). Plants follow a reduced-form (S, s) adjustment rule: capital adjusts to the productivity-implied target $k^*_{j,t} = z_{j,t}^{1/(1-\alpha)}$ if $(k^*_{j,t} - k_{j,t-1})^2 > \theta_i \cdot k_{j,t-1}$; otherwise $k_{j,t} = (1-\delta)k_{j,t-1}$ with $\delta = 0.0083$/month ($\approx 10\%$ annual). Industry-aggregate output is $y_{i,t} = \sum_j z_{j,t} k^\alpha_{j,t}$ with $\alpha = 0.30$.

Calibration of θ_i Industry-specific fixed-cost curvatures are drawn from the cross-industry distribution of Cooper and Haltiwanger (2006) plant-level estimates (Table 4-5), aggregated to NAICS-3 by capital-intensity bands. Values range from $\theta = 0.008$ (Apparel-type) to $\theta = 0.048$ (Petroleum), capturing the empirically observed range of plant-level adjustment-cost dispersion.

Two horse-race tests We run both tests across 50 independent simulation replications:

Test 1: Cross-industry dispersion $\sigma(\ln \omega_i)$. KT generates median $\sigma(\ln \omega_i) = 0.31$ with 95% range $[0.21, 0.41]$. The observed value across the N=16 manufacturing cross-section is 0.36. Sixteen percent of KT replications match or exceed observed dispersion. By this test alone, KT is a viable alternative.

Test 2: φ-channel slope β_γ in the regression of $\ln \omega_i$ on $\ln \gamma_i$ (the Ellison-Glaeser concentration index of the data). KT predicts $\beta_\gamma \approx 0$: capital-adjustment-cost curvature θ_i has no structural relationship to geographic concentration γ_i, so the simulated cross-section produces no γ-ω slope. The KT-implied $\hat{\beta}_\gamma^{\mathrm{KT}}$ across replications has mean 0.02, median 0.02, 95% range $[-0.11, 0.11]$. The empirical slope estimates fall in the $\hat{\beta}_\gamma \in [0.41, 0.48]$ range across multitaper, Burg AR(12), and IV specifications. **Zero of 50 KT replications generate slopes in this range.**

Verdict The KT model can roughly match cross-industry dispersion in cycle frequencies but does not generate the structural relationship between geographic concentration and cycle frequency observed in the data. The φ-channel slope is the discriminating signature: timing search predicts $\beta_\gamma = 1/2$ from primitives; the KT specification considered here predicts $\beta_\gamma = 0$ regardless of θ heterogeneity. The data deliver $\hat{\beta}_\gamma = 0.45$, structurally consistent with timing search and outside the range generated by any of the KT replications considered. The comparison clarifies what additional structure the empirical question requires rather than ruling out enriched versions of these models.

References

Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012): "The Network Origins of Aggregate Fluctuations," *Econometrica*, 80, 1977–2016.

Bartlett, M. S. (1946): "On the Theoretical Specification and Sampling Properties of Autocorrelated Time-Series," *Supplement to the Journal of the Royal Statistical Society*, 8, 27–41.

Beaudry, P., D. Galizia, and F. Portier (2020): "Putting the Cycle Back into Business Cycle Analysis," *American Economic Review*, 110, 1–47.

Burdett, K. and D. T. Mortensen (1998): "Wage Differentials, Employer Size, and Unemployment," *International Economic Review*, 39, 257–273.

Cooper, R. W. and J. C. Haltiwanger (2006): "On the Nature of Capital Adjustment Costs," *Review of Economic Studies*, 73, 611–633.

Ellison, G. and E. L. Glaeser (1997): "Geographic Concentration in U.S. Manufacturing Industries: A Dartboard Approach," *Journal of Political Economy*, 105, 889–927.

Ellison, G., E. L. Glaeser, and W. R. Kerr (2010): "What Causes Industry Agglomeration? Evidence from Coagglomeration Patterns," *American Economic Review*, 100, 1195–1213.

Fisher, R. A. (1929): "Tests of Significance in Harmonic Analysis," *Proceedings of the Royal Society of London. Series A*, 125, 54–59.

Gabaix, X. (2011): "The Granular Origins of Aggregate Fluctuations," *Econometrica*, 79, 733–772.

Hotelling, H. (1929): "Stability in Competition," *Economic Journal*, 39, 41–57.

Khan, A. and J. K. Thomas (2008): "Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Investment Dynamics," *Econometrica*, 76, 395–436.

Kydland, F. E. and E. C. Prescott (1982): "Time to Build and Aggregate Fluctuations," *Econometrica*, 50, 1345–1370.

Lewbel, A. (2012): "Using Heteroscedasticity to Identify and Estimate Mismeasured and Endogenous Regressor Models," *Journal of Business & Economic Statistics*, 30, 67–80.

Ohanian, L. E. (2001): "Why Did Productivity Fall So Much During the Great Depression?" *American Economic Review*, 91, 34–38.

Salop, S. C. (1979): "Monopolistic Competition with Outside Goods," *Bell Journal of Economics*, 10, 141–156.